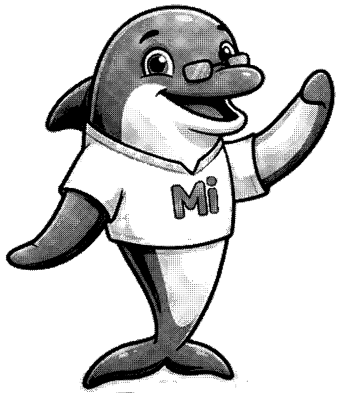
Math Inquirers

# Extension Workbook

## with answers

## Level 3

Self-study | Ages 8-10




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 3**
Self-study practice with answers.


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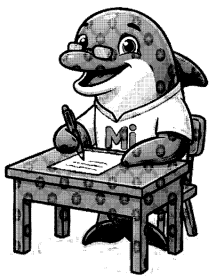
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- Printed copies are black and white. The soft copy has the same pages in colour.

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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

### How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

### For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- You will need: a pencil, a rubber, a ruler and a paper corner.

### The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Digits, cards and clues | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 2 | Find the digit | 13-14 | 15-16 | 17 and 19 | 18 | ☐ |
| 3 | Times table tricks | 20-21 | 22-23 | 24 and 26 | 25 | ☐ |
| 4 | Draw the problem | 27-28 | 29-30 | 31 and 33 | 32 | ☐ |
| 5 | What is left over? | 34-35 | 36-37 | 38 and 40 | 39 | ☐ |
| 6 | Build a wall | 41-42 | 43-44 | 45 and 47 | 46 | ☐ |
| 7 | Around and inside | 48-49 | 50-51 | 52 and 54 | 53 | ☐ |
| 8 | Corners and lines | 55-56 | 57-58 | 59 and 61 | 60 | ☐ |
| 9 | Measure it, pay it | 62-63 | 64-65 | 66 and 68 | 67 | ☐ |
| 10 | Think it through | 69-70 | 71-72 | 73 and 75 | 74 | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

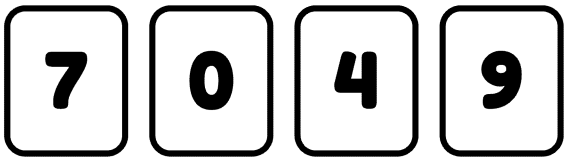
# Learn it: Numbers to 10 000

**My goal.** I can read, build and order 4-digit numbers. I can use clues to find a mystery number.

A 4-digit number has thousands, hundreds, tens and ones. The place decides what a digit is worth. A 0 can never be the first digit.

## Worked example

**Use the cards 7, 0, 4 and 9 once each. Make the greatest and the smallest 4-digit number.**




1. **Greatest: put the greatest digit in the thousands place. Then the next greatest. This gives 9740.**
2. **Smallest: the first digit cannot be 0. Use the smallest digit that is not 0: the 4.**
3. **Put the other digits in order after it, smallest first: 0, 7, 9. This gives 4079.**

**Answer: Greatest 9740. Smallest 4079.**

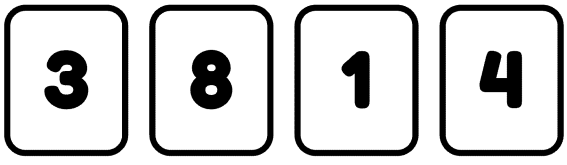
## How to make a number from cards

1. Th H T O: the place decides the worth. 1000 more changes the thousands digit.
2. Odd or even? Choose the ones digit first.
3. Making a list? Fix the first digit, then list the rest from smallest.

## Your turn

Answer at the back of the book.

Try **Make the greatest even number with the cards 3, 8, 1 and 4. Use each card once.** ☐



## Start here

Do these first. Take your time.

**1** **Look at the place value chart.**

| Th | H | T | O |
|---|---|---|---|
| 4 | 0 | 7 | 2 |

(a) Write the number. ☐

(b) The digit 7 is worth ☐.

**2** **Write each number in digits.**

(a) 6 thousands, 5 hundreds and 9 ones ☐

(b) 2 thousands and 15 tens ☐

(c) 5 thousands, 12 hundreds and 3 ones ☐

**3** **Fill in the blanks.**

(a) 1000 more than 6457 is ☐.

(b) 100 less than 3050 is ☐.

(c) 10 more than 8995 is ☐.

(d) 1 less than 7000 is ☐.

**4** **Arrange the numbers in order. Begin with the smallest.**
**5830, 5083, 5380, 5038**

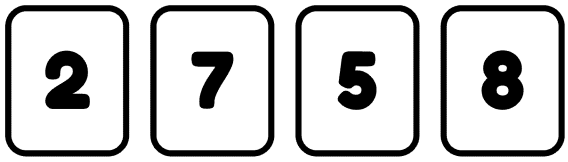
My order: _____

**5** **A shop has 3 boxes of 1000 pins, 14 packs of 100 pins and 6 loose pins. How many pins are there in all?** ☐

## Start here

Keep going. Check your work.

**6** **Use the cards once each in every number.**



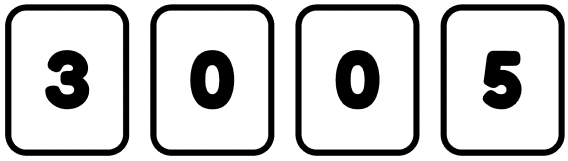

(a) The greatest number ☐

(b) The smallest number ☐

(c) The greatest odd number ☐

(d) The smallest odd number ☐

**7** **List every 4-digit number that can be made with the cards 3, 0, 0 and 5. Begin with the smallest.**




My list: _____

**8** **Write the next 2 numbers in each pattern.**

(a) 2300, 2600, 2900, ☐, ☐

(b) 9000, 8750, 8500, ☐, ☐

(c) 6, 14, 22, 30, ☐, ☐

**9** **A pattern starts at 3000 and goes down by 250 each time.**

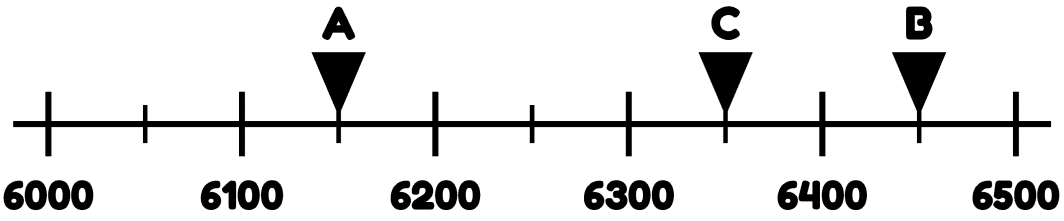
(a) Is 1900 in the pattern? Yes No

(b) The first number below 1900 in the pattern is ☐.

## Start here

Nearly there. You can do it.

**10** **Look at the number line.**




(a) Arrow A points to ☐.

(b) Arrow B points to ☐.

(c) Arrow C points to ☐.

**11** **I am a 4-digit number. My thousands digit is 6, my tens digit is 0, and my hundreds digit is 3 less than 6. My ones digit is the greatest odd digit. What number am I?** ☐

**12** **Ravi says the greatest odd number from the cards 9, 4, 6 and 2 is 9642.**

| 9 | 4 | 6 | 2 |
|---|---|---|---|

(a) Is he right? Yes No

(b) The greatest odd number from the cards is ☐.

**13** **Look at the number 5737.**

(a) The first 7 is worth ☐.

(b) The second 7 is worth ☐.

(c) The first 7 is worth ☐ more than the second 7.

## Think harder

Hints are upside down. Try first!

**1** **The digits of a 4-digit number are all different and add up to 8. The number is even.**

**(a)** The greatest it can be is ☐.

**(b)** The smallest it can be is ☐.

Sets of digits: _____

Numbers I tried: _____

_____

**Hint.** Find the sets of 4 different digits that add up to 8 first.

**2** **Count the 3-digit numbers whose digits multiply to make 8. No digit is 0.**

☐

Sets of digits: _____

Numbers: _____

_____

**Hint.** List the sets of 3 digits first. Then arrange each set.

**3** **Mei makes 4-digit numbers with the cards 1, 2, 3 and 4. She uses each card once.**

**(a)** ☐ **of the numbers are greater than 2500.**

**(b)** The 7th smallest number is ☐.

My list: _____

_____

_____

**Hint.** List them from the smallest. Fix the first digit.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Counting digit sums

**1** **Count the 3-digit numbers with a digit sum of 4. List them by first digit.**

**(a)** Digit sum 4: ☐ numbers.

**(b)** Use the pattern. Digit sum 5: ☐ numbers.

First digit 4: _____

First digit 3: _____

**Hint.** Fix the first digit. See how the counts grow: 1, 2, 3, ...

★ Beyond Level 3: Growing jumps

**2** **Look at the jumps between the numbers. Write the next 2 numbers.**

**5, 8, 14, 23, 35, ☐, ☐**

Jumps: _____

**Hint.** Write the jumps between the numbers. What do you see?

**3** **Make up your own number riddle. Choose a secret number with 4 digits, then write 3 clues. Test your clues: exactly ONE number must fit. Swap riddles with a grown-up.**

Clue 1: _____

Clue 2: _____

Clue 3: _____

**Hint.** Pick the number first. Then write clues about its digits.

**4** **Find the secret number. It is greater than 3000 and less than 4000. Its digits are all different and add up to 15. The tens digit is 0, and the hundreds digit is double the ones digit.** ☐

**Hint.** Try ones digit 1, 2, 3 and 4. Check each sum.

## ✔ Check yourself

No hints. Try each one on your own.

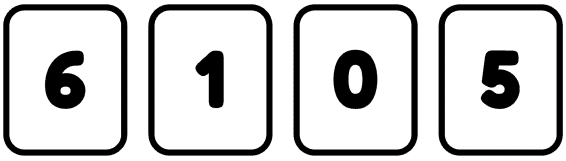
**1** In 7258, the digit 2 is worth ☐.

Working: _____

**2** Write the next 2 numbers. 6500, 6000, 5500, ☐, ☐

Working: _____

**3** Make the smallest 4-digit number from the cards 6, 1, 0 and 5. Use each card once. ☐




**4** Ali has 5 boxes of 1000 beads, 12 bags of 100 beads and 8 loose beads. How many beads does he have? ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Addition and subtraction

**My goal.** I can find missing digits in sums and take-aways. I can solve magic squares and number pyramids.

In a missing-digit sum, work one column at a time. Start with the ones. Remember to add the digit you carried.

## Worked example

**Find the 2 missing digits in this sum.**

| | Th | H | T | O |
|---|---|---|---|---|
| | 4 | 5 | □ | 7 |
| + | 1 | □ | 6 | 8 |
| | 5 | 8 | 9 | 5 |

1. Ones: 7 + 8 = 15. Write 5 and carry 1.
2. Tens: ? + 6 + 1 carried must end in 9. So ? is 2.
3. Hundreds: 5 + ? makes 8. So ? is 3.
4. Check the thousands: 4 + 1 = 5. It matches.

**Answer: 4527 + 1368 = 5895. The missing digits are 2 and 3.**

## How to find a missing digit

1. Look at the ones column first.
2. Write the carried 1 above the next column.
3. For a take-away, turn it round and add to check.

## Your turn

Answer at the back of the book.

**Try** Find the missing digit. □

3?48 + 2517 = 5965

## Start here

Do these first. Take your time.

**1** **Find the missing digit in each sum.**

(a) 3?46 + 1251 = 4597 The missing digit is ☐.

(b) 26?9 + 1435 = 4104 The missing digit is ☐.

**2** **Find the missing digit in each take-away.**

(a) 6405 − 2?18 = 3687 The missing digit is ☐.

(b) 9?30 − 4617 = 4513 The missing digit is ☐.

**3** **A theatre sells 3875 tickets on Saturday and 2968 tickets on Sunday.**

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

**4** **A baker has 8000 g of flour. He uses 2475 g on Monday and 3150 g on Tuesday. How many grams of flour are left?** ☐ g

**5** **Siti works out 5000 − 1768 = 3342.**

(a) Check by adding: 3342 + 1768 = ☐

(b) Is Siti right? Yes No

(c) The correct answer is ☐.

**6** **Look for a pattern.**

(a) 2641 + ☐ = 9999

(b) 10 000 − 2641 = ☐

## Start here

Keep going. Check your work.

**7** **Every row, column and diagonal has the same total.**

(a) The magic total is ☐.

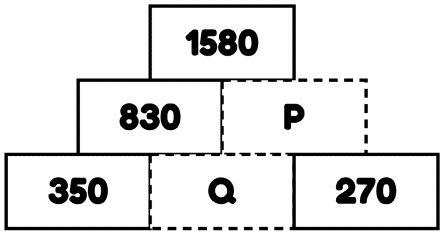
(b) A = ☐ B = ☐ C = ☐

| | | |
|---|---|---|
| 200 | 450 | A |
| 550 | 350 | 150 |
| 300 | B | C |

**8** **A brick is made by adding the 2 bricks below it.**

(a) P = ☐

(b) Q = ☐




**9** **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number being added is ☐.

| | Th | H | T | O |
|---|---|---|---|---|
| | 3 | ☐ | 5 | 2 |
| + | 2 | 4 | ☐ | 6 |
| | 6 | 1 | 4 | 8 |

**10** **Each letter stands for a digit. The same letter always means the same digit. Find the digit.**

(a) A3 + 2A = 56 A = ☐

(b) C5 + C5 = 90 C = ☐

## Start here

Nearly there. You can do it.

**11** **Find the 2 missing digits.**

(a) The digit in the hundreds place of the top number is ☐.

(b) The digit in the tens place of the number taken away is ☐.

| Th | H | T | O |
|---|---|---|---|
| 6 | ☐ | 0 | 2 |
| − 2 | 8 | ☐ | 5 |
| 3 | 6 | 5 | 7 |

**12** **The sum of 2 numbers is 9000. One number is 3475. What is the other number?** ☐

Working: _____

**13** **Use the cards once each to make two 3-digit numbers.**

| 2 | 3 | 4 | 5 | 7 | 8 |
|---|---|---|---|---|---|

(a) The greatest sum is ☐.

(b) The smallest sum is ☐.

**14** **A school collects cans for 3 weeks: 1875, 2430 and 1790.**

(a) In all, the school collects ☐ cans.

(b) Week 2 has ☐ more cans than Week 3.

Working: _____

## Think harder

Hints are upside down. Try first!

**1** **Letters hide digits. The same letter is the same digit. Different letters never share a digit. Find the digits.**

**AB + AC = BBD**

**(a)** A = ☐ **(b)** B = ☐ **(c)** C = ☐ **(d)** D = ☐

My tries: _____

_____

**Hint.** Look at the left digit of the answer. Which digit can it be?

**2** **Use the cards 1, 3, 5, 6 and 8 once each. Make a 3-digit number and a 2-digit number. What is the smallest sum you can make?** ☐

| 1 | 3 | 5 | 6 | 8 |
|---|---|---|---|---|

My tries: _____

_____

_____

**Hint.** Which digits are worth the least? Which places are worth most?

★ **Beyond Level 3: Magic square total**

**3** **Look at this magic square. Every row, column and diagonal has the same total.**

| | | |
|---|---|---|
| 120 | 270 | A |
| | 210 | |
| 180 | | E |

**(a)** The magic total is ☐.

**(b)** A = ☐

**(c)** E = ☐

**Hint.** The middle number is in 4 lines. Add those 4 lines.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Letters hide digits. The same letter is the same digit. Different letters never share a digit. Find the digits.**

**ABC + BC = CAB**

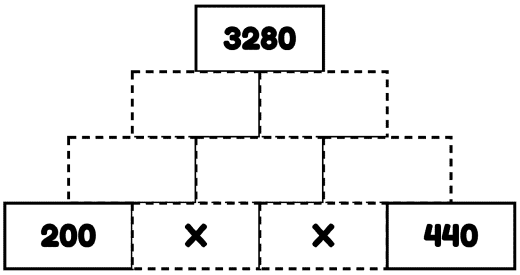
**(a)** A = ☐ **(b)** B = ☐ **(c)** C = ☐

My tries: _____

**Hint.** Look at the ones: C + C ends in B. List the choices for C.

★ **Beyond Level 3: Pyramids with a hidden number**

**2** **A brick is made by adding the 2 bricks below it. The 2 bricks marked X are equal. Find X.** ☐




**Hint.** Guess a number for X. Fill the pyramid. Is the top too big?

★ **Beyond Level 3: Reversing digits**

**3** **Reverse the digits of a 2-digit number and add the 2 numbers.**

**(a)** 36 + 63 = ☐ **(b)** 58 + 85 = ☐ **(c)** 47 + 74 = ☐

**(d)** Each answer is in the ☐ times table.

**Hint.** Add the tens digit and ones digit. Compare with the answers.

**4** **Make up a number pyramid puzzle of your own. The bottom row has 3 bricks. Show exactly 3 bricks and hide the other 3, so that only ONE answer fits. Swap puzzles with a grown-up.**

Draw my pyramid:

**Hint.** Make a full pyramid first. Then hide 3 bricks.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Find the missing digit.** ☐

**4?72 + 2519 = 7091**

Working: _____

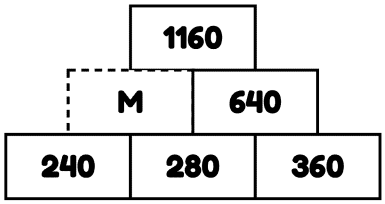
**2** **Find the missing digit.** ☐

**8?04 − 3569 = 4635**

Working: _____

**3** **A brick is made by adding the 2 bricks below it.**

**The brick M is** ☐ **.**




**4** **A zoo has 9000 visitors in 3 days. On Friday 2845 visitors come. On Saturday 3260 come. How many come on Sunday?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

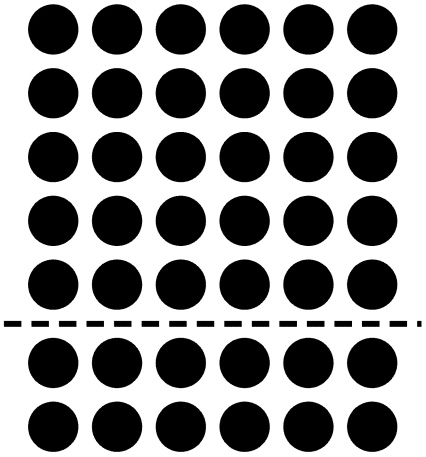
# Learn it: Tables of 6, 7, 8 and 9

**My goal. I can use facts I know to find new times table facts. I can find patterns and solve number puzzles.**

A times table fact you do not know can be built from facts you do know. Split a rectangle of dots into two smaller rectangles.

## Worked example

**There are 7 rows with 6 dots in each row. How many dots are there in all?**



1. **The dashed line splits the dots into 5 rows and 2 rows.**
2. **5 × 6 = 30 and 2 × 6 = 12.**
3. **Add the 2 parts: 30 + 12 = 42.**
4. **So 7 × 6 = 42.**

**Answer: 7 × 6 = 42. There are 42 dots.**

## Ways to find a fact

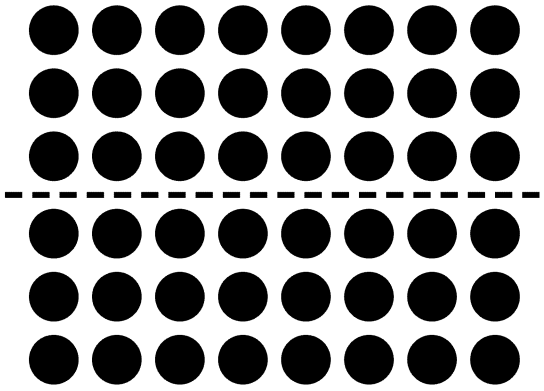
1. Split into facts you know: 5 ×, 2 ×, 10 ×.
2. For 9 ×: take 1 group away from 10 groups.
3. Check with the last digit.

## Your turn

Answer at the back of the book.

Try

**There are 6 rows with 8 dots in each row. Split them into 3 rows and 3 rows. How many dots are there in all?**



## Start here

Do these first. Take your time.

**1** **Write the missing answers in the 8 times table.**

| n | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 × n | 8 | 16 | 24 | | 40 | | 56 | 64 | | 80 |

**2** **Write the answers.**

(a) 6 × 8 = ☐ (b) 7 × 9 = ☐ (c) 9 × 6 = ☐

(d) 8 × 7 = ☐

**3** **You know that 8 × 5 = 40. Use it to find the other facts.**

(a) 8 × 6 = ☐

(b) 8 × 4 = ☐

(c) 8 × 10 = ☐

(d) 8 × 9 = ☐

**4** **Ali packs 7 eggs in each box. He fills 9 boxes.**

(a) He packs ☐ eggs.

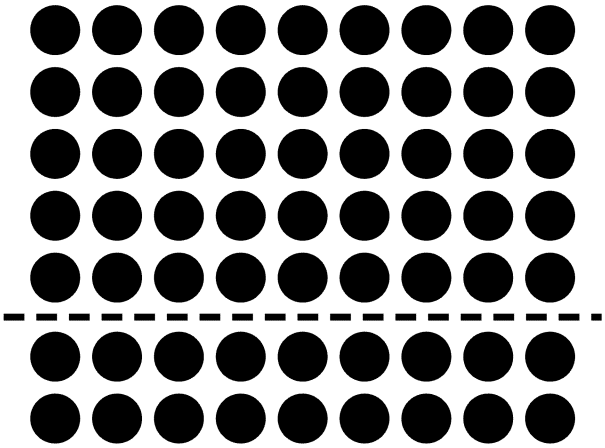
(b) Then he packs 56 eggs into boxes of 7. He fills ☐ boxes.

**5** **Find my number. It is in the 7 times table, and it is greater than 50 and less than 80. Its digits add up to 9.** ☐

## Start here

Keep going. Check your work.

**6** **The dashed line splits 7 rows of 9 dots into 2 parts.**



(a) The top part has 5 × 9 = ☐ dots.

(b) The bottom part has 2 × 9 = ☐ dots.

(c) There are ☐ dots in all.

**7** **Fill in the missing answers.**

| × | 6 | 7 | 8 | 9 |
|---|---|---|---|---|
| 6 | 36 | 42 | A | 54 |
| 7 | B | 49 | 56 | C |
| 8 | 48 | D | 64 | 72 |

A = ☐ B = ☐ C = ☐ D = ☐

**8** **A row number times a column number makes each product. Write the missing numbers in the heading cells.**

| × | | |
|---|---|---|
| | 54 | 63 |
| | 48 | 56 |

## Start here

Nearly there. You can do it.

**9** **Write the last digit of each answer. Do not find the whole answer.**

(a) 6 × 13 ☐

(b) 8 × 27 ☐

(c) 9 × 45 ☐

(d) 7 × 68 ☐

**10** **Mei puts 30 tiles in equal rows. There are at least 2 rows and at least 2 tiles in each row. 2 rows of 15 and 15 rows of 2 are different ways. List ALL the ways.**

There are ☐ ways.

| rows | tiles in each row |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |

**11** **Which of these numbers are in the 6 times table?**
**24, 38, 42, 56, 60**

_____

**12** **Mei buys 6 packs of 8 stickers. Ben buys 7 packs of 7 stickers.**

(a) Who has more stickers? Mei Ben

(b) How many more? ☐

## Think harder

Hints are upside down. Try first!

**1** **Aisha says the 8 times table from 8 up to 96. How many of the numbers she says have an ODD tens digit?** ☐

The 8 times table: _____

_____

Odd tens digit: _____

_____

**Hint.** Write the table first. Then look at the tens digits.

**2** **The row numbers and column numbers are all from 6 to 9. Find the empty cells A to D. You can write in the heading cells to help you.**

(a) A = ☐ (b) B = ☐

(c) C = ☐ (d) D = ☐

| × | | | |
|---|---|---|---|
| | 42 | 48 | A |
| | 49 | B | 63 |
| | C | D | 72 |

**Hint.** Start with 49. Which row and column make 49?

**3** **Two numbers have a product of 63 and a sum of 16. What are the 2 numbers?** ☐ **and** ☐

Pairs that multiply to 63: _____

_____

_____

**Hint.** List the pairs of numbers that multiply to make 63.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Digit sums and the 9s

**1** **Add the digits again and again: 684 gives 6 + 8 + 4 = 18, then 1 + 8 = 9. Numbers in the 9 times table always end at 9. Which of these are in it?**
**738, 825, 963, 1089**

_____

**Hint.** Add the digits of each number. Keep adding until 1 digit is left.

★ Beyond Level 3: Adding every row

**2** **A hall has 8 rows. Row 1 has 7 seats. Each row has 2 more seats than the row before.**

| Row | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Seats | 7 | 9 | | | | | | |

(a) Row 8 has ☐ seats.

(b) The hall has ☐ seats in all.

**Hint.** Pair the first and last rows. Then the next pair.

**3** **Make your own 2 by 2 grid puzzle with numbers from 6 to 9. Show 3 products. Only ONE answer may fit. Swap it with a grown-up.**

_____

**Hint.** Pick the row and column numbers first. Then multiply.

★ Beyond Level 3: Square numbers

**4** **Multiply a number by itself. For example 7 × 7 = 49, which ends in 9. Which of these digits can NEVER end such an answer?**

**2 3 4 6 7 8**

Last digits of 1 × 1 to 9 × 9: _____

**Hint.** Find 1 × 1, 2 × 2, … 9 × 9. Write the last digits.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A box holds 8 pens. How many pens are in 7 boxes?** ☐

Working: _____

**2** **Fill in the blank: 7 × 8 = 7 × 4 + 7 ×** ☐

Working: _____

**3** **Is 63 in the 6 times table?** **Yes** **No**

Working: _____

**4** **Mei has 54 apples. She puts 6 apples in each bag. How many bags does she need?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

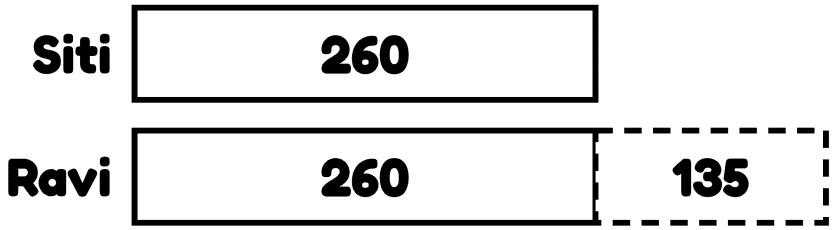
# Learn it: Word problems with bar models

**My goal. I can draw a bar model for a word problem. I can use it to decide what to add, subtract or divide.**

A bar model is a picture of a word problem. Each bar is a length that shows an amount. Longer bar, greater amount.

## Worked example

**Siti has 260 beads. Ravi has 135 more beads than Siti. How many beads do they have in all?**




1. **Draw a bar for Siti. Draw a longer bar for Ravi. The extra part is 135.**
2. **Ravi has 260 + 135 = 395 beads.**
3. **Both children: 260 + 395 = 655 beads.**

**Answer: They have 655 beads in all.**

## How to draw a bar model

1. Draw one bar for each amount.
2. Write what you know on the bars. Put a ? on what you want to find.
3. Look at the bars. Then add, subtract or divide.

## More models

- 3 times as many, 48 in all: 4 boxes, 48 ÷ 4 = 12.
- 50 in all, 10 more: 50 - 10 = 40, so 20 and 30.
- 70 and 50: the gap is 20. Give half the gap, 10: 60 each.

## Your turn

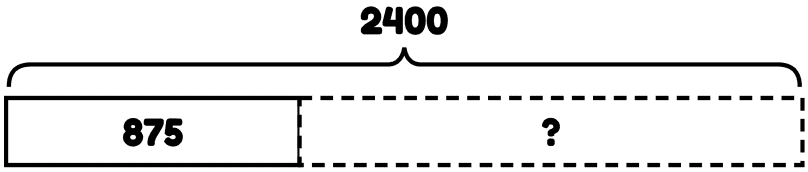
Answer at the back of the book.

Try **Ravi's plant is 480 mm tall. Mei's plant is 215 mm shorter. How tall is Mei's plant? ____ mm**

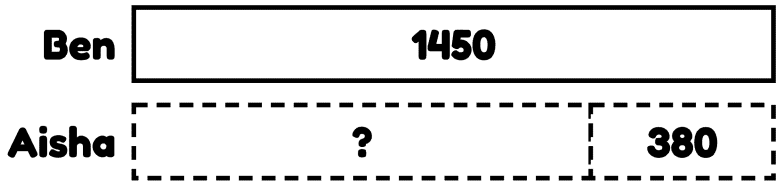
## Start here

Do these first. Take your time.

**1** A farm has 2400 eggs. 875 of them are brown. The rest are white. How many eggs are white? ☐




**2** Ben has 1450 cards. Aisha has 380 fewer cards than Ben.




(a) Aisha has ☐ cards.

(b) They have ☐ cards in all.

**3** A school sells 468 tickets on Friday, 539 on Saturday and 392 on Sunday.

(a) It sells ☐ tickets in all.

(b) It sells ☐ more tickets on Saturday than on Sunday.

**4** Siti has 64 beads. Wei has 29 beads. Each child gets 12 more beads.

(a) Before, Siti has ☐ more beads than Wei.

(b) After, Siti has ☐ more beads than Wei.

## Start here

Keep going. Check your work.

**5** **Ravi and Aisha have 140 coins in all. Ravi has 3 times as many coins as Aisha.**

**(a)** Aisha has ☐ coins.

**(b)** Ravi has ☐ coins.

Draw your model:

Working: _____

_____

**6** **Kumar and Wei have 90 stickers in all. Kumar has 18 more stickers than Wei.**

**(a)** Wei has ☐ stickers.

**(b)** Kumar has ☐ stickers.

Draw your model:

Working: _____

_____

**7** **Ravi's ribbon is 78 cm long. Mei's ribbon is 52 cm long. Ravi cuts a piece off his ribbon and gives it to Mei. Now both ribbons are the same length. How long is the piece?** ☐ **cm**

Draw your model:

Working: _____

## Start here

Nearly there. You can do it.

**8** **Ravi reads 85 pages on Monday. On Tuesday he reads 28 pages more than he does on Monday. On Wednesday he reads 17 pages fewer than on Tuesday.**

(a) He reads ☐ pages on Tuesday.

(b) He reads ☐ pages on Wednesday.

(c) He reads ☐ pages in the 3 days.

**9** **Mei has $50. She buys a book for $18.40 and a pen for $6.75. How much money does she have left? $** ☐

**10** **A zoo has 3 times as many monkeys as parrots. There are 48 monkeys.**

(a) There are ☐ parrots.

(b) There are ☐ monkeys and parrots in all.

Draw your model:

**11** **A shop has 54 red pens. It has 18 fewer blue pens than red pens. Kumar says the shop has 54 + 18 = 72 blue pens.**

(a) Is Kumar right? Yes No

(b) The shop has ☐ blue pens.

**12** **Siti saves $12 each week for 7 weeks. Then she spends $35. How much money does she have left? $** ☐

Working: _____

## Think harder

Hints are upside down. Try first!

★ Beyond Level 3: Giving to make them equal

**1** **A library has 84 books on 2 shelves. The top shelf has 3 times as many books as the bottom shelf. The librarian moves some books from the top shelf to the bottom shelf. Now both shelves have the same number of books.**

**How many books does she move?** ☐

Draw your model:

Working: _____

**Hint.** Find how many each shelf has at first. Then how many later.

**2** **Ali buys a pencil box and 2 pens for $14. Mei buys the same box and 5 pens for $23.**

**(a)** 1 pen costs $ ☐ .

**(b)** The pencil box costs $ ☐ .

Draw your model:

Working: _____

_____

**Hint.** What is different between the 2 bills?

★ Beyond Level 3: Ages and gaps

**3** **Ravi is 7 years old. His father is 35 years old. In how many years will his father be twice as old as Ravi?** ☐ **years**

Draw your model:

Working: _____

**Hint.** The age gap never changes. Draw a bar for each age.

## ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Aisha, Kumar and Lily have 140 stickers in all. Kumar has 12 more stickers than Lily. Aisha has twice as many stickers as Kumar.**

**(a)** Lily has ☐ stickers. **(b)** Kumar has ☐ stickers.

**(c)** Aisha has ☐ stickers.

Draw your model:

**Hint.** Draw Lily as 1 bar. Build Kumar and Aisha from that bar.

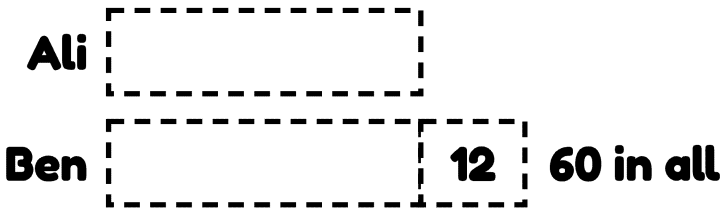
★ **Beyond Level 3: One amount changes**

**2** **A shop has 4 times as many red balloons as blue balloons. It sells 18 red balloons. Now the shop has the same number of red and blue balloons. How many balloons did the shop have at first?** ☐

Draw your model:

**Hint.** Draw blue as 1 box and red as 4 boxes. Selling 18 leaves 1 box.

**3** **Look at the bar model. Make up a story that fits it. Then solve your story.**




My story: _____

_____

My answer: _____

**Hint.** Pick things that fit: stickers, coins, cards or beads.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A library has 2000 books. 735 books are lent out. 640 books are being mended. How many books are on the shelves?** ☐

Working: _____

**2** **Ravi has 560 beads. Siti has 135 more beads than Ravi. How many beads do they have in all?** ☐

Working: _____

**3** **Siti and Kumar have 72 stamps in all. Siti has 5 times the number of stamps that Kumar has. How many stamps does Siti have?** ☐

Working: _____

**4** **Two boxes hold 50 pens in all. Box A has 8 more pens than Box B. How many pens are in Box A?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

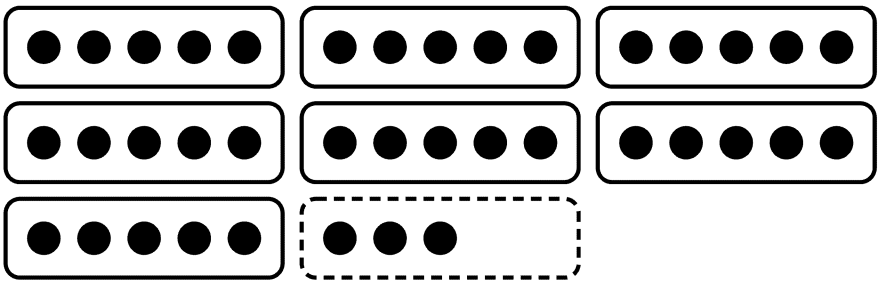
# Learn it: Division with remainders

**My goal. I can divide with a remainder. I can decide what the remainder means in a story. I can find a mystery number.**

When we share or make groups, some may be left over. The leftover is the remainder. It is always smaller than the number we divide by.

## Worked example

**Ali has 38 stickers. Each bag takes 5 stickers. How many bags can he fill? How many stickers are left over?**



1. **Find the greatest fact that is not more than 38. 5 × 7 = 35.**
2. **Subtract to find what is left: 38 − 35 = 3.**
3. **Write it as 38 ÷ 5 = 7 remainder 3.**
4. **Check: 7 × 5 + 3 = 38.**

**Answer: 7 full bags, with 3 stickers left over.**

## How to divide with a remainder

1. Find the greatest table fact that fits in the number.
2. Subtract it to find the remainder.
3. Check: groups × size + remainder = start.
4. A leftover person still needs a seat: use 1 more group.

## Round up or not?

- 31 children, 4 at each table: 31 ÷ 4 = 7 remainder 3. The 3 left over need a table too: 8 tables.
- $31, pens at $4: 7 pens and $3 is left. The leftover is not enough for another pen.

## Your turn

Answer at the back of the book.

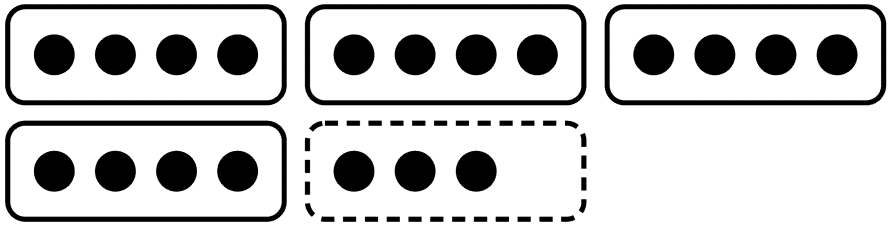
Try **A baker packs 43 buns into boxes of 6. She fills ☐ boxes and has ☐ buns left over.**

Name _____

## Start here

Do these first. Take your time.

**1** **The dots are put into groups of 4.**



(a) There are ☐ full groups. (b) There are ☐ dots left over.

**2** **Divide.**

(a) 38 ÷ 7 = ☐ remainder ☐

(b) 50 ÷ 8 = ☐ remainder ☐

(c) 61 ÷ 9 = ☐ remainder ☐

(d) 75 ÷ 6 = ☐ remainder ☐

**3** **Divide. Then decide what the leftover means.**

(a) 26 people go in cars of 5. They need ☐ cars.

(b) 37 cm of ribbon, cut in 6 cm pieces, gives ☐ whole pieces.

(c) Ravi has $30. A toy costs $4. He buys ☐ toys.

(d) Then he has $ ☐ left.

**4** **Kumar says 59 ÷ 7 = 7 remainder 10.**

(a) The remainder 10 is greater than ☐.

(b) The right answer is ☐ remainder ☐.

 Name _____

## Start here

Keep going. Check your work.

**5** **Lily thinks of a number. She divides it by 6. The answer is 8 remainder 5.**

(a) Lily's number is ☐.

(b) She divides the same number by 7. The answer is ☐ remainder ☐.

**6** **Think about remainders.**

(a) List every remainder that is possible when a number is divided by 5.

_____

(b) The greatest 2-digit number that leaves remainder 5 when divided by 8 is ☐.

**7** **Look for a pattern. Write the remainder when each number is divided by 4.**

| Number | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
|---|---|---|---|---|---|---|---|---|
| Remainder | | | | | | | | |

(a) Fill in the empty cells.

(b) When 30 is divided by 4 the remainder is ☐.

(c) When 100 is divided by 4 the remainder is ☐.

**8** **Find a number between 30 and 40. Dividing it by 6 gives remainder 1. Dividing it by 4 gives remainder 3.**

(a) The numbers that leave remainder 1 when divided by 6 are ☐ and ☐.

(b) The number I am looking for is ☐.

## Start here

Nearly there. You can do it.

**9** **52 children go on a trip. Each bus has 9 seats.**

(a) The children need ☐ buses.

(b) The last bus has ☐ children.

(c) ☐ seats on the last bus are empty.

**10** **The letters A, B, C and D repeat in this order.**

| Place | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Letter | A | B | C | D | A | B | C | D |

(a) The 30th letter is ☐.

(b) The 32nd letter is ☐.

**11** **Mei has 8 bags with 9 beads in each bag, and 5 loose beads. She shares all the beads equally among 9 friends.**

(a) Mei has ☐ beads in all.

(b) Each friend gets ☐ beads and ☐ beads are left over.

**12** **Circle each number that has remainder 3 when you divide it by 5.**

18 23 33 41 48

**13** **Divide 47 by each of 4, 5, 6, 7 and 8. Which number gives the greatest remainder?** 4 5 6 7 8

## Think harder

Hints are upside down. Try first!

**1** **Today is Tuesday.**

**(a)** 45 days from today it will be

Mon Tue Wed Thu Fri Sat Sun

**(b)** 20 days ago it was

Mon Tue Wed Thu Fri Sat Sun

Working: _____

_____

**Hint.** A week has 7 days. Divide by 7 and use the remainder.

**2** **Find the 2-digit number. Dividing it by 5 gives remainder 2. Dividing it by 3 gives remainder 1. Dividing it by 4 gives remainder 3.** ☐

My list: _____

Tests: _____

_____

**Hint.** Begin with the clue that gives the fewest numbers to list.

★ **Beyond Level 3: Same answer and remainder**

**3** **Mei divides a number by 9. The whole-number answer and the remainder are the same. Her number is greater than 0 and less than 100.**

**(a)** When both are 4, her number is ☐.

**(b)** The greatest number that works is ☐.

**(c)** There are ☐ numbers that work.

Working: _____

_____

**Hint.** Try 1 remainder 1, then 2 remainder 2. Look for a pattern.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Remainders when you add and double

**1** **A number leaves remainder 3 when divided by 7.**

**(a)** Add 5 to it. The new remainder is ☐.

**(b)** Double it. The new remainder is ☐.

**(c)** Another number leaves 5 when divided by 8. Doubled, it leaves ☐.

My test numbers: _____

**Hint.** Try a number like 10 (remainder 3 when divided by 7). Test it.

**2** **Ali has between 20 and 100 marbles. Counting in 3s leaves 1 over. Counting in 4s leaves 2 over. Counting in 5s leaves 3 over. How many marbles does he have?** ☐

**Hint.** Compare each clue with a full group.

**3** **Make up your own number riddle: a range and 2 remainder clues. Only ONE number may fit. Swap with a grown-up.**

My riddle: _____

The numbers I tested: _____

**Hint.** Pick your number first. Then find its remainders.

★ Beyond Level 3: Making numbers with 2 sizes

**4** **Ben sells eggs in boxes of 3 and boxes of 5. An order must be whole boxes.**

**(a)** Can he fill an order for 11 eggs? Yes No

**(b)** Orders from 1 to 15 he CANNOT fill:

_____

**Hint.** List the orders he can fill. Which numbers are missing?

 Name _____

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Divide. 52 ÷ 8 = ☐ remainder ☐**

Working: _____

**2** **A bus holds 9 people. How many buses are needed for 40 people? ☐**

Working: _____

**3** **Mei packs 25 muffins into boxes of 6. How many muffins are left over?** ☐

Working: _____

**4** **Find a number between 20 and 30. It leaves remainder 2 when divided by 3 and remainder 4 when divided by 5.** ☐

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

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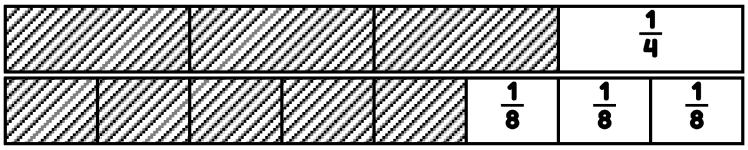
# Learn it: Fractions

**My goal.** I can use a fraction wall to find equal fractions. I can compare fractions and add and subtract them.

A fraction wall shows one whole cut into equal parts. A longer piece of the wall is a greater fraction. To compare fractions, look at where the pieces end.

## Worked example 1

**Which is greater, 5/8 or 3/4? Use the fraction wall.**

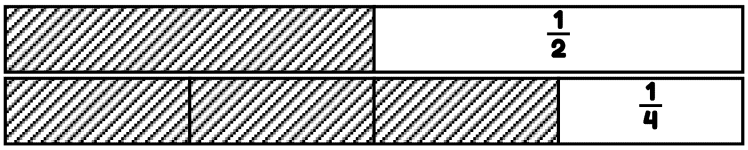



1. Shade 5/8 on the row of eighths.
2. Shade 3/4 on the row of quarters.
3. The end of 3/4 lines up with the end of 6/8.
4. 6/8 is more than 5/8, so 3/4 is greater.

**Answer: 3/4 is greater than 5/8.**

## Worked example 2

**What is 1/2 + 1/4? Use the fraction wall.**




1. On the wall, 1/2 is the same length as 2/4.
2. Add the quarters: 2/4 + 1/4 = 3/4.

**Answer: 1/2 + 1/4 = 3/4.**

## How to compare and add fractions

1. Equal fractions: multiply top and bottom by the same number. 1/3 = 2/6 = 4/12.
2. To add or subtract, change both to the same bottom number.
3. Add or subtract the top numbers. Keep the bottom number.

## Your turn

Answer at the back of the book.

Try **Circle <, = or >. 7/12 < = > 2/3**

## Start here

Do these first. Take your time.

**1** **Use the fraction wall to fill in the blanks.**

| $\frac{1}{3}$ | | $\frac{1}{3}$ | | $\frac{1}{3}$ | |
|---|---|---|---|---|---|
| $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |
| $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ |

(a) 1/3 = ☐ /6 = ☐ /12

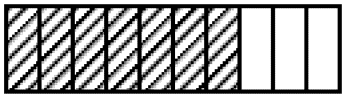
(b) 2/3 = ☐ /6 = ☐ /12

(c) 5/6 = ☐ /12

**2** **Write the fraction of each shape that is shaded.**

A 

B 

C 

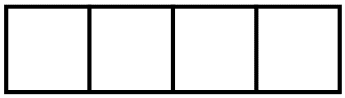
D 

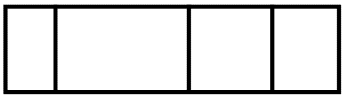
(a) A = ☐ / ☐ (b) B = ☐ / ☐

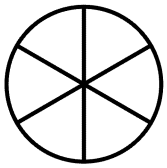
(c) C = ☐ / ☐ (d) D = ☐ / ☐

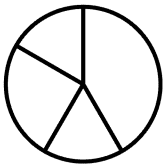
**3** **Which shapes are cut into EQUAL parts? Circle the letters.**

A B C D

A 

B 

C 

D 

Name _____

## Start here

Keep going. Check your work.

**4** Circle <, = or > in each part.

(a) 5/8 < = > 3/8

(b) 1/3 < = > 1/4

(c) 2/4 < = > 4/8

(d) 3/4 < = > 5/6

**5** Arrange the fractions in order. Begin with the smallest.
5/6, 1/3, 1/2, 2/3

_____

**6** Fill in the blanks.

(a) 3/10 + 4/10 = ☐ /10

(b) 5/8 - 2/8 = ☐ /8

(c) 9/12 - 4/12 = ☐ /12

(d) 1 - 3/10 = ☐ /10

**7** Change to the same bottom number. Then fill in the blanks.

(a) 1/2 + 1/4 = ☐ /4

(b) 1/3 + 1/6 = ☐ /6

(c) 3/4 - 1/8 = ☐ /8

(d) 5/6 - 1/2 = ☐ /6

## Start here

Nearly there. You can do it.

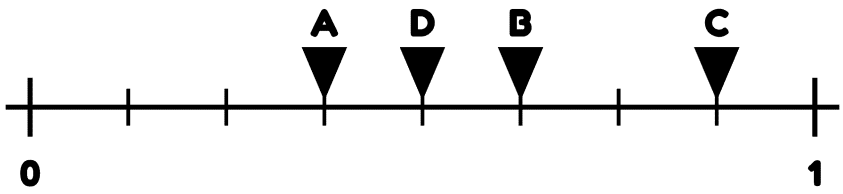
**8** **A cake is cut into 12 equal slices. Ali eats 3 slices. Ben eats 4 slices.**

(a) Ali and Ben eat ☐/12 of the cake.

(b) ☐/12 of the cake is left.

(c) Ben eats ☐/12 more of the cake than Ali.

**9** **Look at the number line from 0 to 1. It is cut into equal steps.**




(a) A = ☐/8

(b) B = ☐/8

(c) C = ☐/8

(d) Which arrow shows 1/2? A B C D

**10** **Fill in the blanks to make equal fractions.**

(a) 3/4 = ☐/12 (b) 2/5 = ☐/10 (c) 1/2 = 5/☐

(d) 6/8 = 3/☐

**11** **Siti says 3/8 + 2/8 = 5/16.**

(a) Is she right? Yes No

(b) The right answer is ☐/8.

## Think harder

Hints are upside down. Try first!

**1** **Look at the fractions with bottom number 12. List every one between 1/3 and 2/3.**

_____

Working: _____

_____

_____

_____

**Hint.** Change 1/3 and 2/3 into twelfths first.

**2** **Ben writes fractions that are equal to 3/4. He adds the top number and the bottom number.**

**(a)** The sum is 28 for the fraction ☐ / ☐.

**(b)** The sum is 42 for the fraction ☐ / ☐.

My list: _____

_____

_____

**Hint.** Write equal fractions of 3/4. Add the top and bottom each time.

**3** **A jug is full of juice. Mei drinks 1/4 of the jug. Ali drinks 1/3 of the jug.**

**(a)** Together they drink ☐ /12 of the jug.

**(b)** ☐ /12 of the jug is left.

Working: _____

_____

_____

**Hint.** Change quarters and thirds into twelfths first.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Adding fractions with top number 1

## 1 Write each as 2 fractions with top number 1.

(a) 3/4 = 1/2 + 1/ ☐ (b) 5/6 = 1/2 + 1/ ☐

(c) 7/12 = 1/3 + 1/ ☐

**Hint.** Take away the first fraction. What is left over?

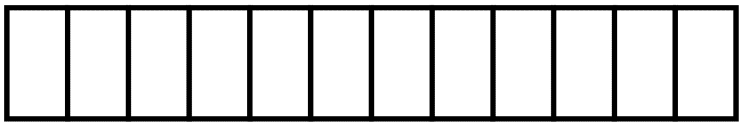
## 2 A whole bar is filled with exactly 6 pieces. The pieces are 1/2, 1/4 or 1/8 of the bar. How many of each?

(a) ☐ halves, ☐ quarters and ☐ eighths.

My tries: _____

**Hint.** Eighths are the smallest. Most of the 6 pieces must be small.

## 3 Colour the bar. Draw lines in 1/2 of the bar. Draw dots in 1/3 of the bar. Leave the rest white.



(a) ☐ parts are white.

(b) That is the same as 1/ ☐ of the bar.

**Hint.** 1/2 of 12 parts is 6 parts. How many parts for 1/3?

## 4 Make up a fraction story about a pizza cut into 8 equal slices. The answer to your story must be 5/8.

My story: _____

My answer: _____

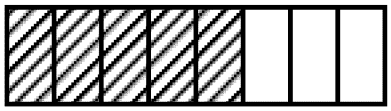
**Hint.** Pick 2 fractions with bottom number 8 that add up to 5/8.

Name _____

## ✔ Check yourself

No hints. Try each one on your own.

**1** Write the fraction that is shaded.



☐/8

**2** Circle <, = or >. 5/8 < = > 3/4

Working: _____

**3** 5/8 − 1/4 = ☐/8

Working: _____

**4** Fill in the blank. 2/3 = ☐/12

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

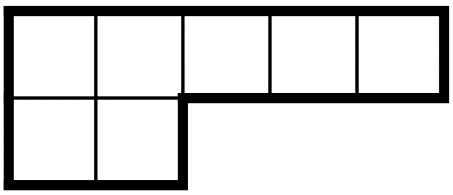
# Learn it: Area and perimeter

**My goal.** I can find the area and the perimeter of rectangles and of shapes made of squares.

Area is the flat space inside a shape. We count squares and write cm². Perimeter is the distance all the way round the edge, in cm.

## Worked example

**The figure is made of squares with sides of 1 cm. What are its area and its perimeter?**



1. **Area: count the squares. 5 + 2 = 7, so the area is 7 cm².**
2. **Perimeter: add the sides all the way round. 5 + 1 + 3 + 1 + 2 + 2 = 14 cm.**
3. **Check: the figure fits in a box 5 cm by 2 cm. 2 × (5 + 2) = 14 cm.**

**Answer: Area 7 cm². Perimeter 14 cm.**

## How to find area and perimeter

1. Area: count the squares, or length × width.
2. Perimeter: add all the sides.
3. Rectangle: perimeter = 2 × (length + width).
4. Write cm for the perimeter and cm² for the area.

## Your turn

Answer at the back of the book.

Try **A rectangle is 9 cm long and 4 cm wide.**

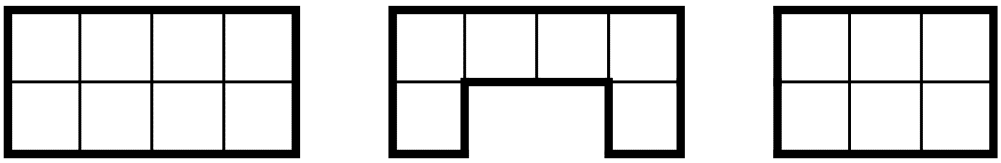
**Its area is** ☐ **cm². Its perimeter is** ☐ **cm.**



## Start here

Do these first. Take your time.

**1** **Every square in the picture has sides of 1 cm. What are the area and the perimeter of each figure?**



A B C

(a) A: area ☐ cm², perimeter ☐ cm

(b) B: area ☐ cm², perimeter ☐ cm

(c) C: area ☐ cm², perimeter ☐ cm

(d) Same area, different perimeter: A,B A,C B,C

**2** **What are the area and the perimeter of each rectangle?**

(a) 8 cm long and 6 cm wide: area ☐ cm², perimeter ☐ cm

(b) 11 m long and 4 m wide: area ☐ m², perimeter ☐ m

**3** **Look at squares.**

(a) A square has sides of 7 cm. Its perimeter is ☐ cm.

(b) Its area is ☐ cm².

(c) A square garden has a fence 36 m long all round. Each side is ☐ m.

**4** **A rectangle has a perimeter of 30 cm. Its length is 9 cm.**

(a) Its width is ☐ cm. (b) Its area is ☐ cm².

## Start here

Keep going. Check your work.

**5** **A mat is 15 cm long and 9 cm wide. Square tiles with sides of 3 cm cover it exactly.**

(a) ☐ tiles fit along the length.

(b) ☐ tiles fit along the width.

(c) The mat needs ☐ tiles in all.

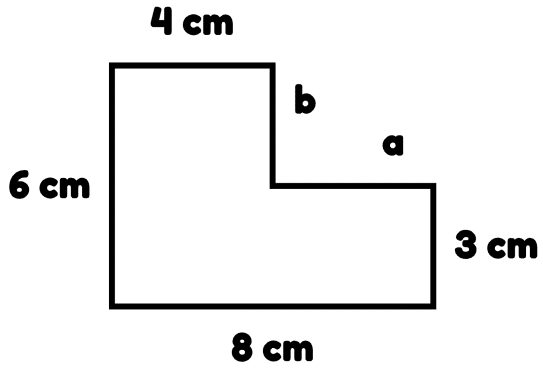
**6** **Mei has 14 cm of wire. She bends all of it into a rectangle. All the sides are whole numbers of cm. List ALL the rectangles.**

(a) There are ☐ different rectangles.

(b) The one with the greatest area has an area of ☐ cm².

| length | width | area |
|---|---|---|
| | | |
| | | |
| | | |
| | | |

**7** **Find the missing sides a and b first. Then find the area, and then the perimeter.**



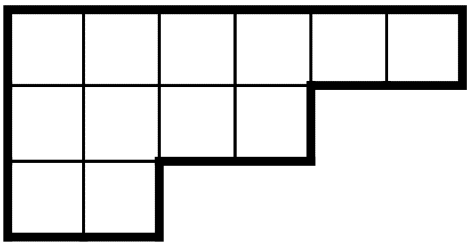

(a) a = ☐ cm (b) b = ☐ cm (c) The area is ☐ cm².

(d) The perimeter is ☐ cm.

## Start here

Nearly there. You can do it.

**8** **Every square in the picture has sides of 1 cm.**



(a) The area is ☐ cm². (b) The perimeter is ☐ cm.

(c) Ben fills the gaps to make a 6 by 3 rectangle. He adds ☐ squares.

(d) The perimeter of the rectangle is ☐ cm.

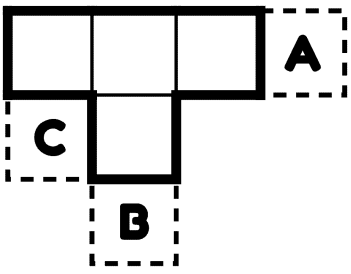
**9** **Mei draws a rectangle 10 cm long and 2 cm wide. Ben draws a square with the same perimeter.**

(a) Each side of Ben's square is ☐ cm.

(b) Whose shape has the greater area? Mei's Ben's

(c) It is greater by ☐ cm².

**10** **The figure has a perimeter of 10 cm. Ali adds ONE square at A, B or C.**




(a) Which places make the perimeter 12 cm? A B C

(b) Which place keeps the perimeter 10 cm?
A B C

**11** **A square room has a floor area of 64 m².**

(a) Each side of the room is ☐ m long.

(b) A border goes all round the floor. It is ☐ m long.

## Think harder

Hints are upside down. Try first!

**1** **A square has sides of 8 cm. One straight cut makes it into 2 equal rectangles.**

**(a)** Each rectangle has a perimeter of ☐ cm.

**(b)** The 2 rectangles together have a perimeter of ☐ cm.

**(c)** That is ☐ cm more than the perimeter of the square.

Working: _____

**Hint.** Draw the square. Draw the cut.

**2** **A rectangle has area 36 cm² and perimeter 26 cm.**

**(a)** Its length is ☐ cm.

**(b)** Its width is ☐ cm.

Pairs that multiply to 36: _____

_____

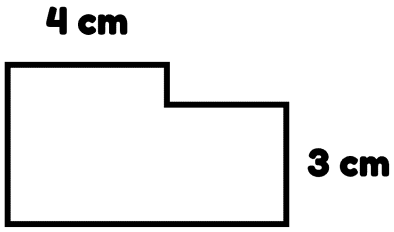
**Hint.** List pairs that multiply to 36. Then add each pair.

**3** **A 4 cm square and a 3 cm square are joined. Their bottom edges are level.**

**(a)** The area of the figure is ☐ cm².

**(b)** The perimeter of the figure is ☐ cm.

**(c)** The 3 cm square is moved on top of the 4 cm square, with their left edges level. The perimeter is then ☐ cm.




**Hint.** Push the steps out to make a box. What is its perimeter?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Perimeter patterns

**1** **Squares with sides of 1 cm are joined in a row. 1 square has a perimeter of 4 cm. 2 squares have 6 cm. 3 squares have 8 cm.**

(a) A row of 10 squares has a perimeter of ☐ cm.

(b) A row with a perimeter of 30 cm has ☐ squares.

My pattern: _____

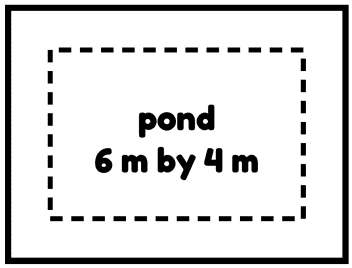
**Hint.** The perimeter goes up by 2 cm for each new square.

★ Beyond Level 3: A path round a pond

**2** **A pond is 6 m long and 4 m wide. A path 1 m wide goes all round the pond.**

(a) Pond and path make a rectangle ☐ m by ☐ m.

(b) The area of the path is ☐ $m^2$.




**Hint.** Find the area of the whole rectangle. Take away the pond.

★ Beyond Level 3: The largest square tile

**3** **A floor is 12 m long and 10 m wide. It is covered exactly with square tiles of one size. The side of a tile is a whole number of metres.**

(a) The largest tile has sides of ☐ m.

(b) The floor needs ☐ of these tiles.

My list: _____

**Hint.** The tile side must fit exactly in 12 and in 10.

## ✔ Check yourself

No hints. Try each one on your own.

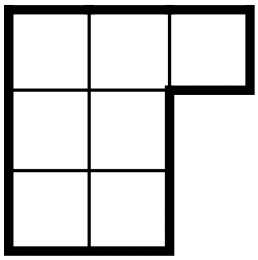
**1** A rectangle is 9 cm long and 5 cm wide. Its area is ☐ cm².

Working: _____

**2** A square tile has a perimeter of 20 cm. Each side is ☐ cm.

Working: _____

**3** Every square has sides of 1 cm. The area is ☐ cm². The perimeter is ☐ cm.



**4** A rectangle has a perimeter of 22 cm. Its width is 4 cm. Its length is ☐ cm.

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

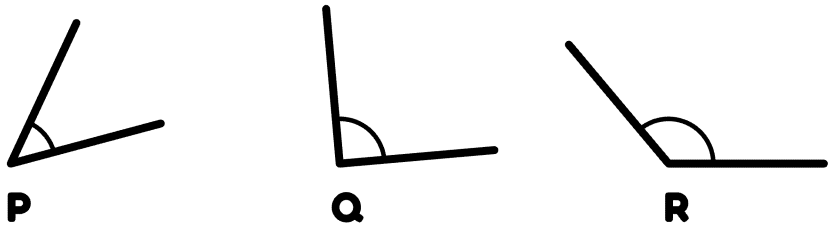
# Learn it: Angles and lines

**My goal. I can compare angles with a right angle. I can find perpendicular and parallel lines. I can count angles in a figure.**

An angle is made where 2 lines meet. A right angle is a square corner, like the corner of a page. Use the corner of a piece of paper as a right angle checker.

## Worked example

**Use a right angle checker. Is each angle smaller than, equal to, or greater than a right angle?**




1. **Put the paper corner on the point where the lines meet.**
2. **Line up one side of the paper with one line.**
3. **Look at the other line. If it is inside the corner, the angle is smaller. If it is outside, the angle is greater.**

**Answer: P is smaller than a right angle. Q is a right angle. R is greater than a right angle.**

## Lines to remember

1. Perpendicular lines meet at a right angle.
2. Parallel lines never meet. They stay the same distance apart.
3. Count angles in order: the small ones first.
4. A corner that points into a shape is a dent. A dent is greater than a right angle.

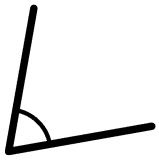
## Your turn

Answer at the back of the book.

Try

**Use a right angle checker. This angle is**

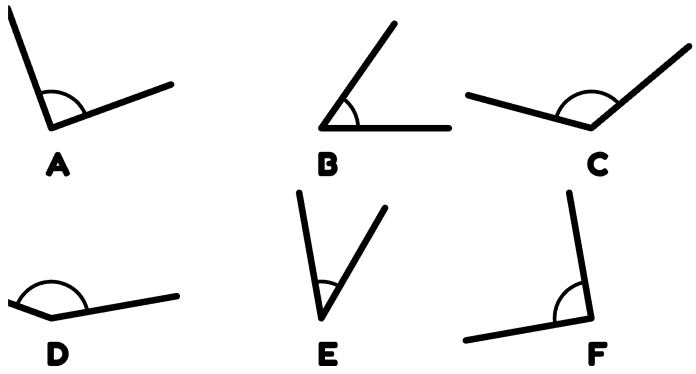
**smaller than equal to greater than a right angle.**



## Start here

Do these first. Take your time.

**1** **Use a right angle checker on each angle. Write the letters.**

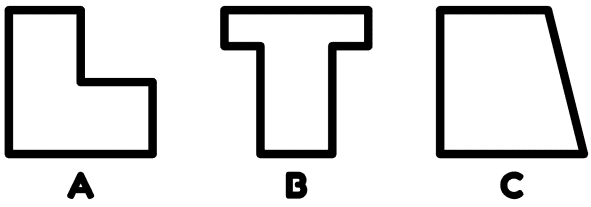



(a) Smaller than a right angle: _____

(b) A right angle: _____

(c) Greater than a right angle: _____

**2** **Count the right angles INSIDE each shape. Check each corner with your paper corner.**




(a) A: ☐ right angles (b) B: ☐ right angles

(c) C: ☐ right angles

**3** **Look at the letters.**

F V N L

(a) Circle each letter with 2 PARALLEL lines. F V N L

(b) Circle each letter with 2 PERPENDICULAR lines. F V N L

## Start here

Keep going. Check your work.

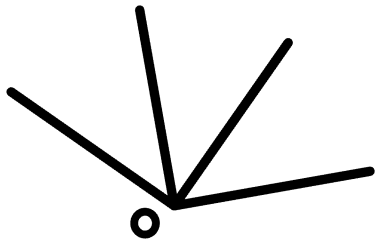
**4** **These 4 lines all start at point O. Count the angles. Count the small ones first.**

(a) Angles made of 1 small angle: ☐

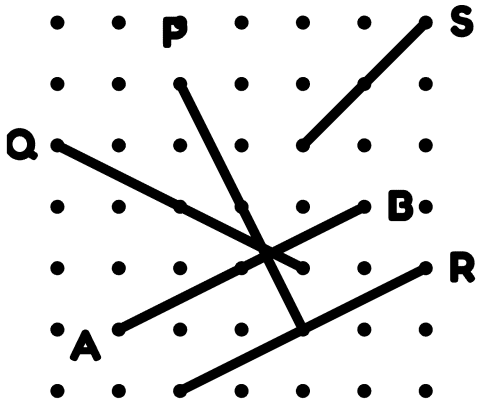
(b) Made of 2 small angles: ☐

(c) Made of 3 small angles: ☐

(d) Angles in all: ☐




**5** **Look at the lines on the dot grid.**



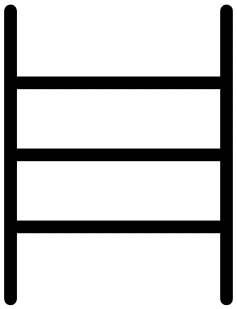

(a) Which line is PARALLEL to line AB? P Q R S

(b) Which line is PERPENDICULAR to line AB? P Q R S

**6** **A ladder has 2 upright rails and 3 rungs. Each rung meets a rail at right angles.**

(a) Where one rung meets the 2 rails there are ☐ right angles.

(b) The ladder has ☐ right angles in all.



## Start here

Nearly there. You can do it.

**7** **Mei faces north. She makes turns of a right angle.**

(a) She turns 1 right angle to the right. She faces N E S W

(b) From north, she turns 3 right angles to the right. She faces
N E S W

(c) From north, she turns 2 right angles to the left. She faces
N E S W

**8** **Fill in the blanks.**

(a) A rectangle has ☐ right angles.

(b) A rectangle has ☐ pairs of parallel sides.

(c) A triangle has at most ☐ right angle.

(d) Two sides that meet at a corner of a square are
parallel perpendicular

**9** **Circle True or False.**

(a) Lines that are perpendicular cross at a right angle. True False

(b) Parallel lines meet if you make them long enough. True False

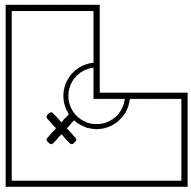
(c) One line can be perpendicular to 2 parallel lines. True False

(d) Two lines that cross are always perpendicular. True False

**10** **Siti says angle X is a right angle.**

(a) Is she right? Yes No

(b) Angle X is
smaller than equal to greater than
a right angle.



## Think harder

Hints are upside down. Try first!

**1** **Lines from a point O make angles. 2 lines give 1 angle, 3 lines give 3 angles and 4 lines give 6 angles.**

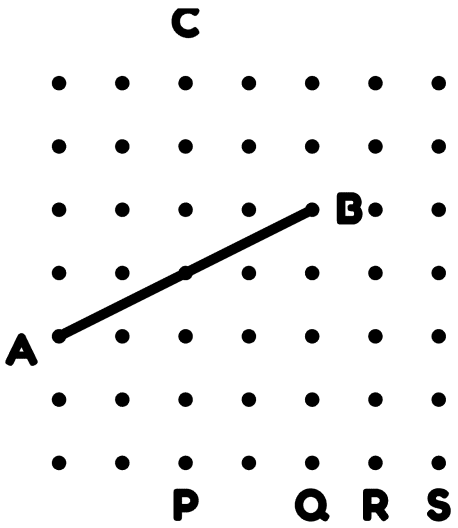
(a) With 5 lines there are ☐ angles.

(b) With 7 lines there are ☐ angles.

Working: _____

**Hint.** Small angles first: 4. Then angles made of 2 small ones: 3.

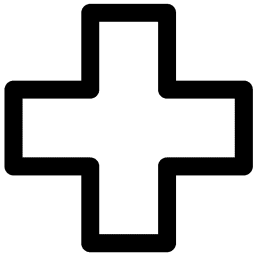
**2** **Draw a line from C that is PERPENDICULAR to line AB. Use a ruler. Make it reach the bottom row of dots.**




(a) Your line reaches the bottom row at P Q R S

**Hint.** AB goes 2 across and 1 up. Turn it a quarter turn.

**3** **Look at the plus sign. All its corners are square.**



(a) It has ☐ right angles inside.

(b) It has ☐ dents.

(c) Another shape with square corners has 10 corners and 3 dents. It has ☐ right angles inside.

**Hint.** Count the dents first. Which corners are left?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: Degrees

**1** **A right angle is 90 degrees.**

(a) 2 right angles make a straight line: ☐ degrees.

(b) 4 right angles make a full turn: ☐ degrees.

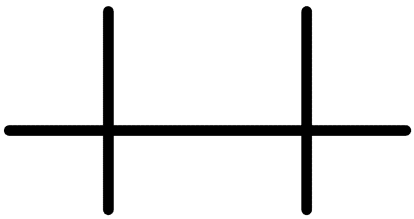
(c) Half of a right angle is ☐ degrees.

(d) 3 right angles make ☐ degrees.

**Hint.** Add the right angles: 90 + 90 + ...

★ Beyond Level 3: Lines and a third line

**2** **Two upright lines both cross the long line at right angles.**



(a) The 2 upright lines are parallel perpendicular

(b) A line is perpendicular to one of 2 parallel lines. Is it also perpendicular to the other? Yes No

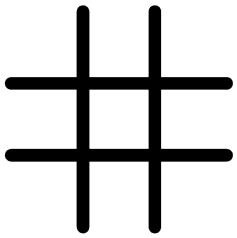
**Hint.** Do the 2 upright lines ever meet?

**3** **A board has 2 lines across and 2 lines up.**

(a) There are ☐ pairs of parallel lines.

(b) The lines make ☐ right angles where they cross.

(c) A board with 3 lines across and 3 lines up makes ☐ right angles.



**Hint.** Each place where 2 lines cross makes 4 right angles.

## ✔ Check yourself

No hints. Try each one on your own.

**1**

**Use a right angle checker. This angle is**

**smaller than equal to greater than a right angle.**



**2** **Circle each letter that has parallel lines.** **E H V T**

E H V T

**3** **A rectangle has ☐ pairs of parallel sides.**

Working: _____

**4** **Ben faces east. He turns 1 right angle to the left. He faces**

**N E S W**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

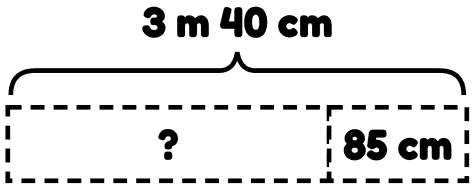
# Learn it: Length, mass, volume, money and time

**My goal.** I can change units, solve money problems and find how long things last.

Use one unit before you add or take away. 1 m = 100 cm, 1 kg = 1000 g, 1 ℓ = 1000 ml and $1 = 100¢.

## Worked example 1

**A ribbon is 3 m 40 cm long. Mei cuts off 85 cm. How long is the ribbon now?**

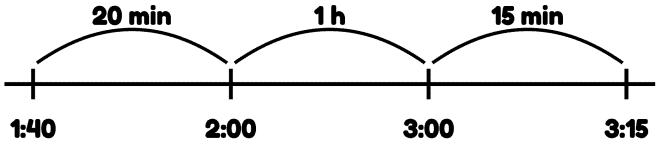



1. Change to cm: 3 m 40 cm = 340 cm.
2. Take away: 340 − 85 = 255 cm = 2 m 55 cm.

**Answer: The ribbon is 2 m 55 cm long.**

## Worked example 2

**A film starts at 1:40 pm and ends at 3:15 pm. How long is the film?**




1. Jump to the next o'clock, then to 3:00, then to 3:15.
2. Add the jumps: 20 min + 1 h + 15 min = 1 h 35 min.

**Answer: The film is 1 h 35 min long.**

## Tips

1. Write both amounts in the small unit first.
2. On a time line, jump to the next o'clock.

## Your turn

Answer at the back of the book.

Try **A jug holds 2 ℓ 350 ml. Ali pours out 800 ml. The jug now holds** ☐ ℓ ☐ **ml.**

## Start here

Do these first. Take your time.

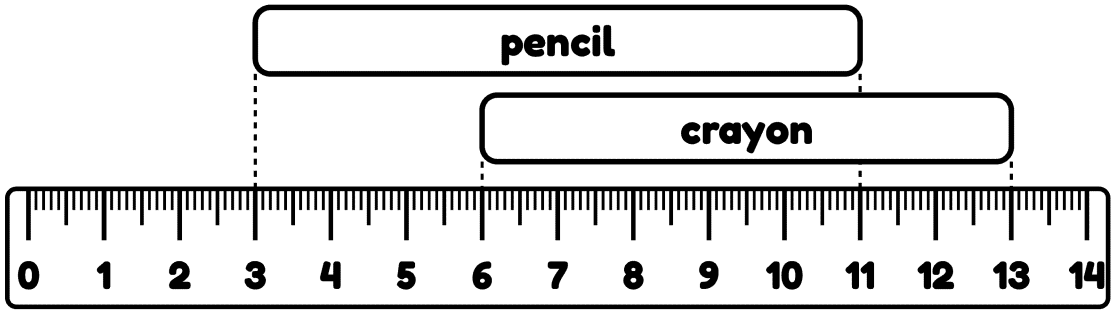
**1** **Change the units.**

(a) 3 m = ☐ cm (b) 4 kg 50 g = ☐ g

(c) 5 ℓ 300 ml = ☐ ml (d) 2750 g = ☐ kg ☐ g

(e) 365 cm = ☐ m ☐ cm

**2** **The ruler is shown at real size. Read the length of each object.**




(a) The pencil is ☐ cm long.

(b) The crayon is ☐ cm long.

(c) The pencil is ☐ cm longer than the crayon.

**3** **The parcels weigh: A 1 kg 80 g, B 1900 g, C 1 kg 8 g, D 1 kg 90 g. Write the letters in order. Begin with the lightest.**

☐ ☐ ☐ ☐

**4** **A can has 3 ℓ 200 ml of paint. Siti pours the paint into pots. Each pot holds 500 ml.**

(a) She fills ☐ pots.

(b) ☐ ml of paint are left.

## Start here

Keep going. Check your work.

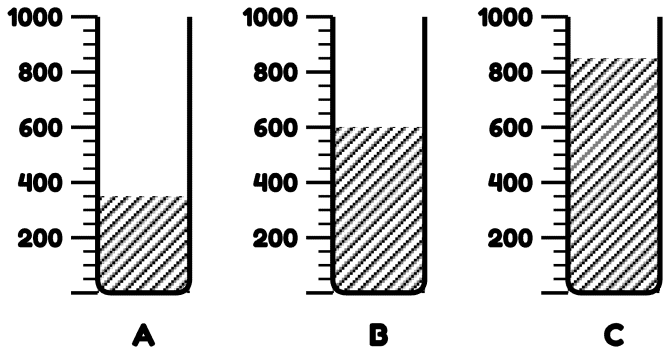
**5** **Read the water level in each jug.**

(a) A holds ☐ ml

(b) B holds ☐ ml

(c) C holds ☐ ml

(d) C has ☐ ml more than A




**6** **A notebook costs $2.65 and a ruler costs $1.40. Mei pays with a $5 note.**

(a) They cost $ ☐ in all.

(b) Her change is $ ☐.

**7** **Siti has $20 and buys the first 3 things on the list.**

| Book | Puzzle | Bottle | Stickers |
|---|---|---|---|
| $6.20 | $4.85 | $3.40 | $1.75 |

(a) They cost $ ☐ in all.

(b) Her change is $ ☐.

(c) Can she buy all 4 things on the list? Yes No

**8** **3 melons with the same mass weigh 2 kg 400 g. An apple weighs 150 g. 1 melon and 2 apples weigh ☐ g.**

Working: _____

## Start here

Nearly there. You can do it.

**9** **Write the time in the other kind of time.**

(a) 6:15 am = ☐ (b) 1:40 pm = ☐

(c) 9:05 pm = ☐ (d) 16:50 = ☐ pm

(e) 20:25 = ☐ pm

**10** **How long does each take? Use a time line to help.**

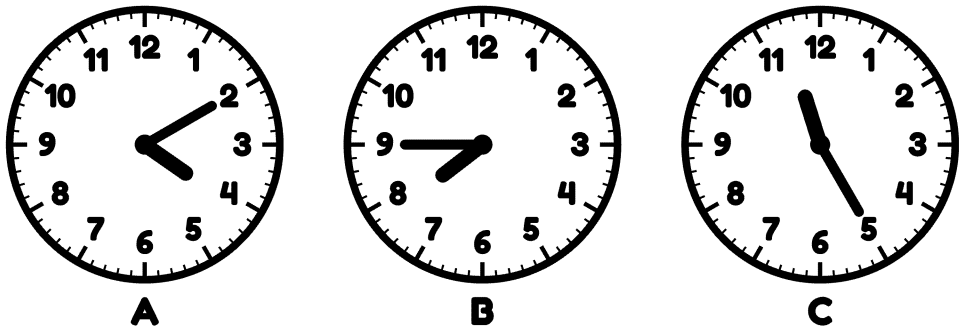
(a) Aisha's class: 5:25 pm to 6:40 pm. It lasts ☐ h ☐ min.

(b) Ben's class starts at 3:40 pm and lasts 55 minutes. It ends at ☐.

(c) A show ends at 8:15 pm. It is 1 h 45 min long. It starts at ☐.

Time line: _____

**11** **Write the time shown on each clock.**




(a) A: ☐ (b) B: ☐ (c) C: ☐

**12** **A stamp costs 65¢. Kumar buys 8 stamps.**

(a) The stamps cost $ ☐.

(b) He pays with a $10 note. His change is $ ☐.

## Think harder

Hints are upside down. Try first!

**1** **A pool has 4 swimming lessons in a row. Each lesson is 45 minutes. There is a 10 minute break between lessons. The first lesson starts at 07:30. When does the 4th lesson end?** ☐

Lesson 1: _____

Lesson 2: _____

Lesson 3: _____

_____

**Hint.** Write the start and the end of each lesson.

**2** **Ali starts his homework at 18:35 and stops at 20:10. He has a 15 minute break. How long does he work?** ☐ **h** ☐ **min**

Time line: _____

_____

**Hint.** Find the whole time first. Then take away the break.

★ **Beyond Level 3: Two bills in one**

**3** **2 notebooks and 3 pens cost $9. A notebook costs $1.50 more than a pen.**

**(a)** A pen costs $ ☐.

**(b)** A notebook costs $ ☐.

Working: _____

_____

_____

**Hint.** How much more do 2 notebooks cost than 2 pens?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 3: A water puzzle

**1** **You have a 3 ℓ jug, a 5 ℓ jug and a tap. The jugs have no marks. You want exactly 4 ℓ.**

**(a)** Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

**(b)** Empty the 3 ℓ jug. Pour the 2 ℓ into it. Fill the 5 ℓ jug. Pour from it into the 3 ℓ jug until full. The 5 ℓ jug has ☐ ℓ left.

**Hint.** Follow each step. The jug is full when it holds 3 ℓ.

**2** **Mei's watch is 8 minutes slow.**

**(a)** Her watch shows 4:20 pm. The real time is ☐.

**(b)** A film starts at 4:45 pm. When it starts, her watch shows ☐.

**Hint.** A slow watch shows an earlier time than the real time.

**3** **Write a shopping story that uses $6.40 and $2.15. The answer to your story must be $4.25.**

My story: _____

_____

My answer: _____

**Hint.** Which sum or take-away of $6.40 and $2.15 gives $4.25?

**4** **A night shift starts at 22:30 and lasts 8 h 45 min. At what time does it end?**

**Write it in 24-hour time.** ☐

Time line: _____

**Hint.** How long is it from 22:30 to midnight?

## ✔ Check yourself

No hints. Try each one on your own.

**1** **2 m 40 cm = ☐ cm**

Working: _____

**2** **A bottle holds 1 ℓ 200 ml. A child drinks 700 ml. The bottle then holds ☐ ml.**

Working: _____

**3** **A bus trip starts at 9:40 am and ends at 11:10 am. It lasts ☐ h ☐ min.**

Working: _____

**4** **Kumar buys a snack for $2.35. He pays with a $5 note. His change is $ ☐ .**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

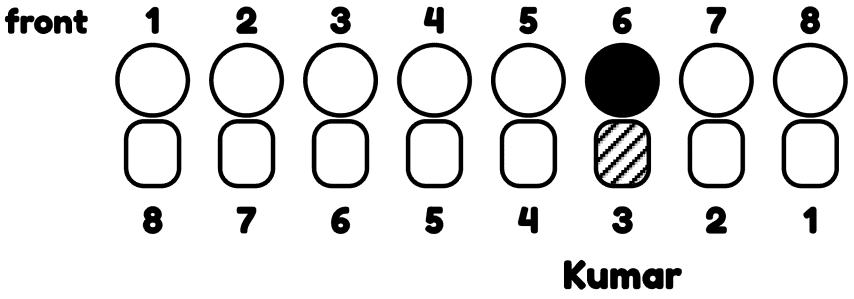
# Learn it: Mixed puzzles

**My goal.** I can start a puzzle by drawing it, listing it or making a table. I can read a bar graph and solve logic puzzles.

To start a puzzle, try one of 3 ways. Draw it, list it, or make a table. Then check every clue.

## Worked example

**Kumar is 6th from the front and 3rd from the back of a line. Count the children in the whole line.**




1. Draw the line. Put Kumar in 6th place from the front.
2. Kumar is 3rd from the back, so 2 children are behind him.
3. 5 children are in front of Kumar, then Kumar, then 2 behind: 5 + 1 + 2 = 8.
4. Check: 6 + 3 − 1 = 8.

**Answer: There are 8 children in the line.**

## Three ways to start

1. Draw it: make a picture of the puzzle.
2. List it: write every case in order.
3. Table: write what you know in rows and columns.
4. Check your answer against every clue.

## Your turn

Answer at the back of the book.

Try **Lily is 3rd from the front and 9th from the back of a line. The line has ☐ children.**

## Start here

Do these first. Take your time.

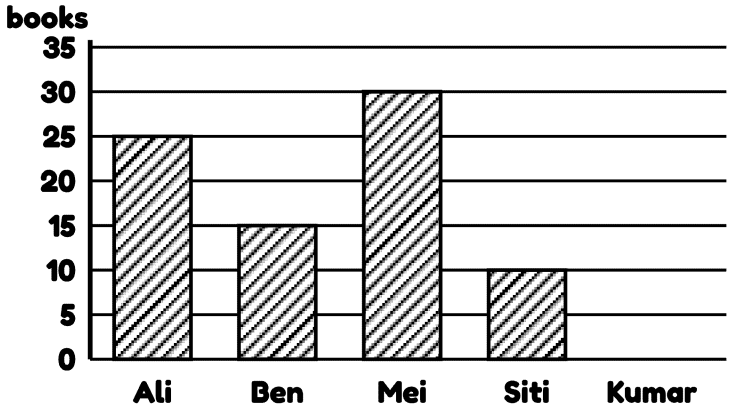
**1** **The bar graph shows the books 4 children read. Kumar read 15 more books than Siti. Draw Kumar's bar on the graph.**

(a) Most books: ☐

(b) Mei read ☐ more than Ben.

(c) Kumar read ☐.

(d) In all they read ☐.



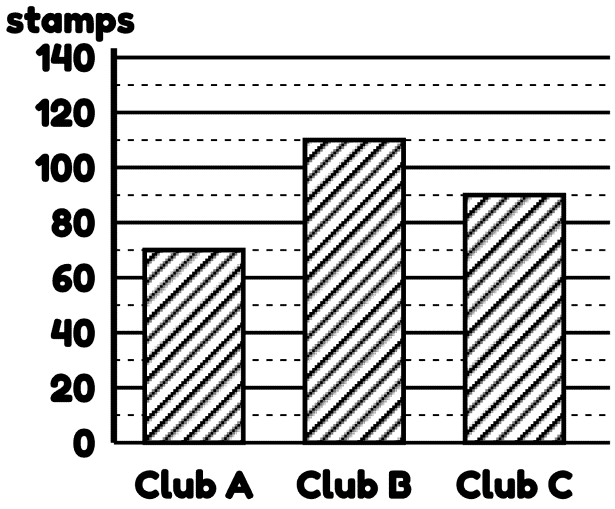

**2** **The graph shows the stamps 3 clubs collected. Each thin line is 10 stamps.**

(a) Club A: ☐

(b) Club B: ☐

(c) Club C: ☐

(d) All 3: ☐




**3** **The 3rd of a month is a Tuesday.**

(a) The 24th is a Mon Tue Wed Thu Fri Sat Sun

(b) The 30th is a Mon Tue Wed Thu Fri Sat Sun

Working: _____

## Start here

Keep going. Check your work.

**4** **Each child has a different fruit. Kumar has neither the apple nor the banana. The child with the orange has a name with 4 letters. Wei does not have the apple. Use ticks and crosses in the grid.**

| | apple | banana | grapes | orange |
|---|---|---|---|---|
| Kumar | | | | |
| Aisha | | | | |
| Wei | | | | |
| Lily | | | | |

(a) Kumar has the apple banana grapes orange

(b) Aisha has the apple banana grapes orange

(c) Wei has the apple banana grapes orange

(d) Lily has the apple banana grapes orange

**5** **Draw it. Count the children in each line.**

(a) Ravi is 3rd from the front. Mei is last. 5 children stand between them. The line has ☐ children.

(b) Aisha is 8th from the front and 3rd from the back. The line has ☐ children.

Draw it:

**6** **A café meal is 1 drink and 1 snack. The drinks are milk and tea. The snacks are a bun, a pie and fruit. List every different meal.**

(a) Write the list below.

(b) There are ☐ meals with milk.

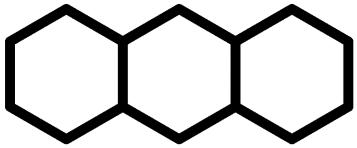
(c) There are ☐ meals with a bun or a pie.

My list: _____

## Start here

Nearly there. You can do it.

**7** **Hexagons are joined in a row with sticks. 1 hexagon needs 6 sticks. 2 hexagons need 11 sticks. 3 hexagons need 16 sticks. Make a table.**



(a) 6 hexagons need ☐ sticks.

(b) 41 sticks make ☐ hexagons.

| hexagons | sticks |
|---|---|
| | |
| | |
| | |
| | |

**8** **Which numbers fit all the clues? The number is even and has 3 digits. It is less than 400. Its digits add up to 9, and none is 0. The tens digit is more than the ones digit.**

_____

**9** **I think of a number. I multiply it by 3 and take away 4. The answer is 17. My number is ☐.**

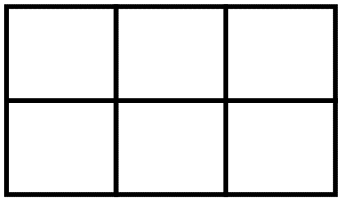
Working backwards: _____

**10** **Aisha has $10. A juice costs $1.85 and a sandwich costs $3.40. She buys 2 juices and 1 sandwich. Her change is $ ☐.**

## Think harder

Hints are upside down. Try first!

**1** **A grid is made of 2 rows of 3 small squares. Count the rectangles of every size, squares included.** ☐



Height 1, width 1, 2, 3: _____

Height 2, width 1, 2, 3: _____

**Hint.** Count the small squares first. Then the wider ones.

**2** **One of Ben, Aisha and Wei broke a cup. They each say 1 thing: Ben "Wei did it", Aisha "Not me", Wei "Ben is lying". Exactly 1 of them tells the truth. Who broke the cup? Ben Aisha Wei**

My tests: _____

_____

_____

**Hint.** Pick one child. Are exactly 1 of the 3 sayings true?

**3** **A shop sold 185 pencils in 4 days. Tuesday sold 15 more than Monday. Wednesday sold twice as many as Monday. Thursday sold 10 fewer than Wednesday.**

**(a)** Monday: ☐ **(b)** Tuesday: ☐ **(c)** Wednesday: ☐

**(d)** Thursday: ☐

Draw your bars:

_____

**Hint.** Draw Monday as 1 bar. Then draw the other days with it.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Count the squares of every size in a 4 by 4 grid. Begin with the small ones.**

(a) 1 by 1 squares: ☐ (b) 2 by 2 squares: ☐

(c) 3 by 3 squares: ☐ (d) All the squares: ☐

**Hint.** A 2 by 2 square can sit in 3 places across and 3 down.

★ Beyond Level 3: A balance puzzle

**2** **9 coins look the same. One is lighter. You have a balance scale.**

(a) With 3 coins, you can always find the light coin in ☐ weighing.

(b) With 9 coins, you can always find it in ☐ weighings.

**Hint.** Put 3 coins on each side. What does the scale tell you?

**3** **Ali has 8 coins. They are 10¢ coins and 20¢ coins, worth $1.50 in all. Find how many coins of each kind.**

(a) ☐ ten-cent coins

(b) ☐ twenty-cent coins

| 10¢ coins | 20¢ coins | value |
|---|---|---|
| | | |
| | | |

**Hint.** Try 4 of each. How much are they worth?

**4** **Make up a logic puzzle with 3 friends and 3 sports. Write 3 clues. Only ONE answer may fit. Swap with a grown-up.**

My clues: _____

_____

_____

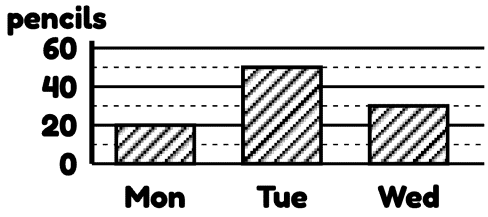
**Hint.** Pick the answer first. Then write clues that rule out the others.

## ✔ Check yourself

No hints. Try each one on your own.

**1**

**Each dashed line is 10. Tuesday sold ☐ more than Monday.**




**2** **In a line, 2 children stand in front of Ben and 6 children stand behind him. The line has ☐ children.**

Working: _____

**3** **Lily is shorter than Mei. Mei is shorter than Ben. Ben is shorter than Aisha. Who is the 2nd tallest?** **Lily** **Mei** **Ben** **Aisha**

Working: _____

**4** **Look for a pattern. 2, 6, 10, 14, … The 8th number is ☐.**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Digits, cards and clues | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Find the digit | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Times table tricks | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Draw the problem | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | What is left over? | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Build a wall | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Around and inside | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Corners and lines | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Measure it, pay it | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Think it through | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

_____

**You finished the Math Inquirers Extension Workbook, Level 3.**

_____ _____

Date Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 1 · Digits, cards and clues

**Your turn**
page 6

**Answer: 8314**
**Why:** An even number ends in 8 or 4. With 4 last, the other cards in order give 8314. With 8 last, the best is 4318. So 8314 is greater.

## START HERE

**1**
page 7

**Answer: (a) 4072 (b) 70**
**Why:** Read the places from the left: 4 thousands, 0 hundreds, 7 tens and 2 ones make 4072. The 7 is in the tens place, so it is worth 70.

**2**
page 7

**Answer: (a) 6509 (b) 2150 (c) 6203**
**Why:** (a) 6000 + 500 + 9 = 6509. (b) 15 tens is 150, so 2000 + 150 = 2150. (c) 12 hundreds is 1200, so 5000 + 1200 + 3 = 6203.

**3**
page 7

**Answer: (a) 7457 (b) 2950 (c) 9005 (d) 6999**
**Why:** (a) Only the thousands digit changes: 7457. (b) 3050 - 100 = 2950. (c) 8995 + 10 = 9005, and the 9 hundreds carry. (d) 7000 - 1 = 6999.

**4**
page 7

**Answer: 5038, 5083, 5380, 5830**
**Why:** All four start with 5 thousands. Compare the hundreds next: 0, 0, 3, 8. Then compare the tens of 5038 and 5083: 3 is less than 8.

**5**
page 7

**Answer: 4406**
**Why:** 3 boxes are 3000 pins. 14 packs are 1400. 3000 + 1400 + 6 = 4406.

**6**
page 8

**Answer: (a) 8752 (b) 2578 (c) 8725 (d) 2587**
**Why:** (a) Greatest digits first: 8752. (b) Smallest first, no 0 here: 2578. (c) The ones digit is 7 or 5. 5 last: 8725 beats 8527. (d) 7 last: 2587 beats 2785.

**7**
page 8

**Answer: 3005, 3050, 3500, 5003, 5030, 5300**
**Why:** The first digit is 3 or 5, never 0. After 3, the 5 goes in the hundreds, tens or ones place: 3005, 3050, 3500. After 5, the same: 5003, 5030, 5300. 6 numbers.

**8**
page 8

**Answer: (a) 3200, 3500 (add 300) (b) 8250, 8000 (subtract 250) (c) 38, 46 (add 8)**
**Why:** (a) Add 300 each time. (b) Subtract 250 each time. (c) Add 8: 38, then 46.

**9**
page 8

**Answer: (a) No (b) 1750**
**Why:** The pattern goes 3000, 2750, 2500, 2250, 2000, 1750. It jumps over 1900 and lands on 1750.

**10**
page 9

**Answer: (a) 6150 (b) 6450 (c) 6350**
**Why:** Each small mark is 50. A is halfway from 6100 to 6200: 6150. B is halfway from 6400 to 6500: 6450. C is halfway from 6300 to 6400: 6350.

**11**
page 9

**Answer: 6309**
**Why:** The thousands digit is 6. The hundreds digit is 6 - 3 = 3. The tens digit is 0. The greatest odd digit is 9. The number is 6309.

**12** page 9
**Answer: (a) No (b) 6429**
**Why:** 9642 ends in 2, so it is even. The only odd card is 9, so 9 must be in the ones place. The other cards in order are 6, 4, 2. The number is 6429.

**13** page 9
**Answer: (a) 700 (b) 7 (c) 693**
**Why:** First 7: hundreds, 700. Second 7: ones, 7. 700 - 7 = 693.

## THINK HARDER

**1** page 10
**Answer: (a) 5210 (b) 1034**
**Why:** Only two sets of 4 different digits add up to 8: 0, 1, 2, 5 and 0, 1, 3, 4. Even numbers need an even last digit. Greatest even: 5210. Smallest: 1034.

**2** page 10
**Answer: 10**
**Why:** The digit sets are 1, 1, 8 and 1, 2, 4 and 2, 2, 2. Set 1, 1, 8 makes 118, 181, 811: 3 numbers. Set 1, 2, 4 makes 6. Set 2, 2, 2 makes 222. 3 + 6 + 1 = 10.

**3** page 10
**Answer: (a) 12 (b) 2134**
**Why:** Numbers that start with 1 or 2 are below 2500, because the second digit is at most 4. Only those that start with 3 or 4 are greater: 6 + 6 = 12. The list begins 1234, 1243, 1324, 1342, 1423, 1432, so the 7th is 2134.

## GO FURTHER

**1** page 11
**Answer: (a) 10 (b) 15**
**Why:** Sum 4: first digit 4 gives 1 number (400), 3 gives 2, 2 gives 3, 1 gives 4: 1 + 2 + 3 + 4 = 10. Sum 5, first digit 1 gives 5 numbers: 1 + 2 + 3 + 4 + 5 = 15.

**2** page 11
**Answer: 50, 68**
**Why:** The jumps are 3, 6, 9, 12. Each jump is 3 more than the one before. The next jumps are 15 and 18: 35 + 15 = 50 and 50 + 18 = 68.

**3** page 11
**Answer: Answers vary. Example: I am between 7000 and 8000. My hundreds digit is double my tens digit. My ones digit is 1 more than my tens digit. My digits add up to 16. The only number is 7423.**
**Why:** List the numbers that fit clue 1, then cross out those that fail clue 2, and so on. In the example, tens digit t gives 7 + 2t + t + (t + 1) = 16, so t = 2 and the number is 7423.

**4** page 11
**Answer: 3804**
**Why:** The thousands digit is 3 and the tens digit is 0. If the ones digit is o, the hundreds digit is 2o, so 3 + 2o + o = 15 and o = 4. The hundreds digit is 8. The number is 3804.

## CHECK YOURSELF

**1** page 12
**Answer: 200**
**Why:** The 2 is in the hundreds place, so it is worth 200.

**2** page 12
**Answer: 5000, 4500**
**Why:** Subtract 500 each time: 5500 - 500 = 5000 and 5000 - 500 = 4500.

**3** page 12
**Answer: 1056**
**Why:** The first digit cannot be 0, so use 1, then 0, 5, 6. The number is 1056.

**4** page 12
**Answer: 6208**
**Why:** 5 boxes are 5000. 12 bags are 1200. 5000 + 1200 + 8 = 6208.

## Unit 2 · Find the digit

**Your turn** page 13
**Answer: 4**
**Why:** Ones: 8 + 7 = 15, write 5 and carry 1. Tens: 4 + 1 + 1 = 6. Hundreds: ? + 5 makes 9, so ? is 4. Thousands: 3 + 2 = 5.

### START HERE

**1** page 14
**Answer: (a) 3 (b) 6**
**Why:** (a) Ones 6 + 1 = 7 and tens 4 + 5 = 9 are fine. Hundreds: ? + 2 = 5, so ? is 3. (b) Ones: 9 + 5 = 14, carry 1. Tens: ? + 3 + 1 ends in 0, so ? is 6.

**2** page 14
**Answer: (a) 7 (b) 1**
**Why:** Turn each one round and add. (a) 3687 + 2718 = 6405, so the digit is 7. (b) 4513 + 4617 = 9130, so the digit is 1.

**3** page 14
**Answer: (a) 6843 (b) 907**
**Why:** (a) 3875 + 2968 = 6843. (b) 3875 - 2968 = 907.

**4** page 14
**Answer: 2375**
**Why:** First add what he used: 2475 + 3150 = 5625. Then 8000 - 5625 = 2375.

**5** page 14
**Answer: (a) 5110 (b) No (c) 3232**
**Why:** The check should give 5000, but 3342 + 1768 = 5110. So Siti is not right. 5000 - 1768 = 3232.

**6** page 14
**Answer: (a) 7358 (b) 7359**
**Why:** (a) Each digit and its partner add up to 9: 2 and 7, 6 and 3, 4 and 5, 1 and 8. That gives 7358. (b) 10 000 is 1 more than 9999, so the answer is 1 more than 7358.

**7** page 15
**Answer: (a) 1050 (b) A = 400, B = 250, C = 500**
**Why:** Middle row: 550 + 350 + 150 = 1050. Top row: 200 + 450 + A = 1050, so A = 400. Middle column: 450 + 350 + B = 1050, so B = 250. Bottom row: 300 + 250 + C = 1050, so C = 500.

**8** page 15
**Answer: (a) 750 (b) 480**
**Why:** 830 + P = 1580, so P = 750. Q + 270 = 750, so Q = 480.

**9** page 15
**Answer: (a) 6 (b) 9**
**Why:** Ones: 2 + 6 = 8. Tens: 5 + ? ends in 4, so ? is 9, carry 1. Hundreds: ? + 4 + 1 ends in 1, so ? is 6, carry 1. Thousands: 3 + 2 + 1 = 6.

**10** page 15
**Answer: (a) 3 (b) 4**
**Why:** (a) Try A = 3: 33 + 23 = 56. It works. (b) Two equal numbers add to 90, so each is 45. Then C = 4.

**11** page 16
**Answer: (a) 5 (b) 4**
**Why:** Turn it round: 3657 + 2845 = 6502. Or subtract by columns: 12 - 5 = 7, 10 - 1 - 4 = 5, 15 - 1 - 8 = 6, 6 - 1 - 2 = 3. The top number is 6502.

**12** page 16
**Answer: 5525**
**Why:** 9000 - 3475 = 5525.

**13** page 16
**Answer: (a) 1595 (b) 605**
**Why:** Greatest sum: the 2 greatest digits go in the hundreds places, the next 2 in the tens, the last 2 in the ones: 800 + 700 + 50 + 40 + 3 + 2 = 1595. Smallest: 200 + 300 + 40 + 50 + 7 + 8 = 605.

**14** page 16
**Answer: (a) 6095 (b) 640**
**Why:** (a) 1875 + 2430 = 4305, 4305 + 1790 = 6095. (b) 2430 - 1790 = 640.

## THINK HARDER

**1** page 17
**Answer: (a) 5 (b) 1 (c) 9 (d) 0**
**Why:** Two 2-digit numbers make less than 200, so B = 1 and the answer starts 11. Ones: 1 + C ends in D. Tens: A + A + carry makes 11, so A = 5 with carry 1. Then 1 + C = 10, so C = 9, D = 0.

**2** page 17
**Answer: 194**
**Why:** The smallest digit, 1, goes in the hundreds place. The next smallest, 3 and 5, go in the tens places. 6 and 8 go in the ones places. For example 136 + 58 = 194, and every such way gives 194.

**3** page 17
**Answer: (a) 630 (b) 240 (c) 300**
**Why:** The middle number is in 4 lines (middle row, middle column, 2 diagonals). Add these 4 lines: every number counts once and the middle counts 3 more times. All the numbers make 3 totals (3 rows). So 4 totals = 3 totals + 3 × the middle, and 1 total = 3 × 210 = 630. Then A = 630 - 120 - 270 = 240 and E = 630 - 120 - 210 = 300.

## GO FURTHER

**1** page 18
**Answer: (a) 2 (b) 6 (c) 3**
**Why:** Ones: C + C ends in B. Tens: B + B (+ carry) ends in A. Hundreds: A (+ carry) is C. Try C = 3: B = 6, carry 0. Tens: 6 + 6 = 12, so A = 2 and carry 1. 2 + 1 = 3 = C. No other digits work.

**2** page 18
**Answer: 440**
**Why:** The end bricks of the bottom row count once in the top brick and the 2 middle bricks 3 times each. So 200 + 6 X + 440 = 3280, 6 X = 2640 and X = 440.

**3** page 18
**Answer: (a) 99 (b) 143 (c) 121 (d) 11**
**Why:** 99 = 9 × 11, 143 = 13 × 11, 121 = 11 × 11. A number with digits A and B is 10 A + B and its reverse is 10 B + A. Together: 11 × (A + B), always in the 11 times table.

**4** page 18
**Answer:** Answers vary. Example: top brick 800, middle left brick 370, bottom right brick 180 shown. The hidden bricks are 430 (middle right), 250 (bottom middle) and 120 (bottom left).
**Why:** In the example: 800 - 370 = 430, 430 - 180 = 250, 370 - 250 = 120.

### CHECK YOURSELF

**1** page 19
**Answer:** 5
**Why:** Ones: 2 + 9 = 11, carry 1. Tens: 7 + 1 + 1 = 9. Hundreds: ? + 5 ends in 0, so ? is 5 and 1 is carried. Thousands: 4 + 2 + 1 = 7.

**2** page 19
**Answer:** 2
**Why:** Turn it round: 4635 + 3569 = 8204, so the missing digit is 2.

**3** page 19
**Answer:** 520
**Why:** M is the sum of 240 and 280: 240 + 280 = 520.

**4** page 19
**Answer:** 2895
**Why:** 2845 + 3260 = 6105. Then 9000 - 6105 = 2895.

## Unit 3 · Times table tricks

**Your turn** page 20
**Answer:** 48
**Why:** 3 × 8 = 24 and 3 × 8 = 24. Then 24 + 24 = 48. So 6 × 8 = 48.

### START HERE

**1** page 21
**Answer:** 32, 48, 72
**Why:** The answers go up by 8. 8 × 4 = 32, 8 × 6 = 48 and 8 × 9 = 72.

**2** page 21
**Answer:** (a) 48 (b) 63 (c) 54 (d) 56
**Why:** 6 × 8 = 48, 7 × 9 = 63, 9 × 6 = 54 and 8 × 7 = 56. Swap the numbers to check: 8 × 6 = 48.

**3** page 21
**Answer:** (a) 48 (b) 32 (c) 80 (d) 72
**Why:** (a) One more 8: 40 + 8 = 48. (b) One less 8: 40 - 8 = 32. (c) Double 8 × 5: 80. (d) 8 × 10 = 80, take away one 8: 72.

**4** page 21
**Answer:** (a) 63 (b) 8
**Why:** (a) 9 × 7 = 63. (b) 56 ÷ 7 = 8, because 7 × 8 = 56.

**5** page 21
**Answer:** 63
**Why:** The 7 times table between 50 and 80: 56, 63, 70, 77. Their digits add up to 11, 9, 7 and 14. Only 63 has digits that add up to 9.

**6** page 22
**Answer:** (a) 45 (b) 18 (c) 63
**Why:** 5 × 9 = 45 and 2 × 9 = 18. Add: 45 + 18 = 63. So 7 × 9 = 63.

**7** page 22
**Answer:** A = 48, B = 42, C = 63, D = 56
**Why:** Each cell is the row number × the column number. A = 6 × 8 = 48. B = 7 × 6 = 42. C = 7 × 9 = 63. D = 8 × 7 = 56.

**8** page 22
**Answer: Columns 6 and 7. Rows 9 and 8.**
**Why:** 54 and 63 are both in the 9 times table: 54 = 9 × 6, 63 = 9 × 7. So the row is 9 and the columns are 6 and 7. Second row: 48 = 8 × 6 and 56 = 8 × 7, so the row is 8.

**9** page 23
**Answer: (a) 8 (b) 6 (c) 5 (d) 6**
**Why:** Only the last digits matter. (a) 6 × 3 = 18 ends in 8. (b) 8 × 7 = 56 ends in 6. (c) 9 × 5 = 45 ends in 5. (d) 7 × 8 = 56 ends in 6.

**10** page 23
**Answer: 6**
**Why:** Try the number of rows from 2 upwards: 2 rows of 15, 3 rows of 10, 5 rows of 6, 6 rows of 5, 10 rows of 3 and 15 rows of 2. 4 rows does not work because 30 is not in the 4 times table. That is 6 ways.

**11** page 23
**Answer: 24, 42, 60**
**Why:** 6 × 4 = 24, 6 × 7 = 42 and 6 × 10 = 60. The numbers 38 and 56 are not: 6 × 6 = 36 and 6 × 9 = 54, and the next answer after 54 is 60.

**12** page 23
**Answer: (a) Ben (b) 1**
**Why:** Mei has 6 × 8 = 48 stickers. Ben has 7 × 7 = 49. Ben has 1 more.

## THINK HARDER

**1** page 24
**Answer: 5**
**Why:** The numbers are 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88 and 96. The tens digits are 0, 1, 2, 3, 4, 4, 5, 6, 7, 8, 8, 9. The odd ones are 1, 3, 5, 7, 9: so 16, 32, 56, 72 and 96. That is 5 numbers.

**2** page 24
**Answer: (a) 54 (b) 56 (c) 56 (d) 64**
**Why:** 49 = 7 × 7, so row 2 and column 1 are 7. 42 = 6 × 7 gives row 1 = 6. 48 = 6 × 8 gives column 2 = 8. 63 = 7 × 9 gives column 3 = 9. 72 = 8 × 9 gives row 3 = 8. A = 54, B = 56, C = 56, D = 64.

**3** page 24
**Answer: 7 and 9**
**Why:** The pairs that multiply to 63 are 1 and 63, 3 and 21, 7 and 9. Their sums are 64, 24 and 16. Only 7 and 9 add up to 16.

## GO FURTHER

**1** page 25
**Answer: 738, 963, 1089**
**Why:** 738: 7 + 3 + 8 = 18, then 1 + 8 = 9. 825: 8 + 2 + 5 = 15, then 1 + 5 = 6, so not. 963: 9 + 6 + 3 = 18, then 9. 1089: 1 + 0 + 8 + 9 = 18, then 9.

**2** page 25
**Answer: (a) 21 (b) 112**
**Why:** The seats go 7, 9, 11, 13, 15, 17, 19, 21. Row 8 has 21 seats. Pair them: 7 + 21 = 28, 9 + 19 = 28, 11 + 17 = 28, 13 + 15 = 28. 4 × 28 = 112 seats.

**3** page 25
**Answer: Answers vary. Example: the products 42, 48 and 49 are shown. The row numbers are 6 and 7 and the column numbers are 7 and 8, so the hidden product is 56.**
**Why:** In the example, 42 = 6 × 7 and 48 = 6 × 8 give row 6 and columns 7 and 8. Also 49 = 7 × 7 gives row 7. The hidden product is 7 × 8 = 56.

**4** page 25
**Answer: 2, 3, 7 and 8**
**Why:** The last digits of 1 × 1 up to 9 × 9 are 1, 4, 9, 6, 5, 6, 9, 4, 1. Only the last digit matters, so a square ends only in 0, 1, 4, 5, 6 or 9. Never 2, 3, 7 or 8.

### CHECK YOURSELF

**1** page 26
**Answer: 56**
**Why:** 7 × 8 = 56.

**2** page 26
**Answer: 4**
**Why:** 8 = 4 + 4, so 7 × 8 = 7 × 4 + 7 × 4. That is 28 + 28 = 56.

**3** page 26
**Answer: No**
**Why:** 6 × 10 = 60 and 6 × 11 = 66, so 63 is not in the table.

**4** page 26
**Answer: 9**
**Why:** 54 ÷ 6 = 9, because 6 × 9 = 54.

## Unit 4 · Draw the problem

**Your turn** page 27
**Answer: 265**
**Why:** Mei's plant is shorter, so subtract: 480 - 215 = 265 mm. Check: 265 + 215 = 480.

### START HERE

**1** page 28
**Answer: 1525**
**Why:** The whole bar is 2400 eggs. One part is 875: 2400 - 875 = 1525.

**2** page 28
**Answer: (a) 1070 (b) 2520**
**Why:** (a) Aisha has fewer: 1450 - 380 = 1070. (b) Together: 1450 + 1070 = 2520.

**3** page 28
**Answer: (a) 1399 (b) 147**
**Why:** (a) 468 + 539 = 1007, and 1007 + 392 = 1399. (b) 539 - 392 = 147.

**4** page 28
**Answer: (a) 35 (b) 35**
**Why:** Before: 64 - 29 = 35. After: Siti has 76 and Wei has 41, and 76 - 41 = 35. The gap stays the same because both get the same number of beads.

**5** page 29
**Answer: (a) 35 (b) 105**
**Why:** There are 4 equal boxes in all. 140 ÷ 4 = 35, so 1 box is 35. Aisha has 1 box: 35 coins. Ravi has 3 boxes: 3 × 35 = 105 coins.

**6** page 29
**Answer: (a) 36 (b) 54**
**Why:** Take away the extra 18 from the total: 90 - 18 = 72. Share the 72 into 2 equal parts: 36 each. Wei has 36. Kumar has 36 + 18 = 54.

**7** page 29
**Answer: 13**
**Why:** The gap is 78 - 52 = 26 cm. The piece closes the gap from both sides, so it is half of the gap: 26 ÷ 2 = 13 cm.

**8** page 30
**Answer: (a) 113 (b) 96 (c) 294**
**Why:** (a) 85 + 28 = 113. (b) 113 - 17 = 96. (c) 85 + 113 + 96 = 294. Wednesday is compared with Tuesday, not with Monday.

**9** page 30
**Answer: 24.85**
**Why:** First add what she spends: 18.40 + 6.75 = 25.15. Then 50.00 - 25.15 = 24.85.

**10** page 30
**Answer: (a) 16 (b) 64**
**Why:** Monkeys are 3 boxes and parrots are 1 box. 3 boxes are 48, so 1 box is 48 ÷ 3 = 16. Together they are 4 boxes: 4 × 16 = 64.

**11** page 30
**Answer: (a) No (b) 36**
**Why:** There are fewer blue pens than red pens, so fewer than 54. Kumar added instead of subtracting. The right answer is 54 - 18 = 36.

**12** page 30
**Answer: 49**
**Why:** In 7 weeks she saves 7 × $12 = $84. Then 84 - 35 = 49: she has $49.

## THINK HARDER

**1** page 31
**Answer: 21**
**Why:** At first there are 4 equal boxes: 84 ÷ 4 = 21, so the bottom shelf has 21 and the top shelf has 63. At the end each has 84 ÷ 2 = 42. She moves 63 - 42 = 21 books.

**2** page 31
**Answer: (a) 3 (b) 8**
**Why:** The second bill has 3 more pens and costs $23 - $14 = $9 more. So 3 pens cost $9 and 1 pen costs $3. The box costs $14 - 2 × $3 = $8.

**3** page 31
**Answer: 21**
**Why:** The gap is 35 - 7 = 28 years and it never changes. When the father is twice as old, Ravi is 1 box and his father is 2 boxes, so 1 box = the gap = 28. Ravi is 28, which is 21 years from now.

## GO FURTHER

**1** page 32
**Answer: (a) 26 (b) 38 (c) 76**
**Why:** Lily is 1 box. Kumar is 1 box and 12. Aisha is 2 boxes and 24. In all: 4 boxes and 36 = 140. So 4 boxes = 104 and 1 box = 26. Lily 26, Kumar 38, Aisha 76.

**2** page 32
**Answer: 30**
**Why:** Red is 4 boxes and blue is 1 box. After selling 18 red, red is 1 box. So 3 boxes are 18 and 1 box is 6. At first: 6 blue and 24 red, 30 balloons.

**3** page 32
**Answer: Answers vary. Example: Ali and Ben have 60 stickers in all. Ben has 12 more stickers than Ali. Ali has 24 stickers and Ben has 36 stickers.**
**Why:** In the example, take away the 12: 60 - 12 = 48, 48 ÷ 2 = 24. Ali has 24 and Ben has 36.

### CHECK YOURSELF

**1** page 33
**Answer: 625**
**Why:** 2000 - 735 = 1265. Then 1265 - 640 = 625.

**2** page 33
**Answer: 1255**
**Why:** Siti has 560 + 135 = 695 beads. Together: 560 + 695 = 1255.

**3** page 33
**Answer: 60**
**Why:** There are 6 equal boxes. 72 ÷ 6 = 12. Siti has 5 boxes: 5 × 12 = 60.

**4** page 33
**Answer: 29**
**Why:** Take away the extra 8: 50 - 8 = 42. Share 42 into 2 equal parts: 21 each. Box B has 21 and Box A has 21 + 8 = 29.

## Unit 5 · What is left over?

**Your turn** page 34
**Answer: 7 full boxes, 1 bun left over**
**Why:** 6 × 7 = 42 and 43 - 42 = 1. So 43 ÷ 6 = 7 remainder 1. Check: 7 × 6 + 1 = 43.

### START HERE

**1** page 35
**Answer: (a) 4 (b) 3**
**Why:** Count the solid boxes: 4 groups of 4 dots make 16 dots. The dashed box has the 3 dots left over. So 19 ÷ 4 = 4 remainder 3.

**2** page 35
**Answer: (a) 5 remainder 3 (b) 6 remainder 2 (c) 6 remainder 7 (d) 12 remainder 3**
**Why:** (a) 7 × 5 = 35 and 38 - 35 = 3. (b) 8 × 6 = 48 and 50 - 48 = 2. (c) 9 × 6 = 54 and 61 - 54 = 7. (d) 6 × 12 = 72 and 75 - 72 = 3.

**3** page 35
**Answer: (a) 6 (b) 6 (c) 7 (d) 2**
**Why:** (a) 26 ÷ 5 = 5 remainder 1. The 1 person left over still needs a car, so 6 cars. (b) 37 ÷ 6 = 6 remainder 1: 6 whole pieces. (c) 30 ÷ 4 = 7 remainder 2: 7 toys. (d) The remainder is the money left: $2.

**4** page 35
**Answer: (a) 7 (b) 8 remainder 3**
**Why:** A remainder must be smaller than the number we divide by. 10 is greater than 7, so another group of 7 fits. 7 × 8 = 56 and 59 - 56 = 3. So 59 ÷ 7 = 8 remainder 3.

**5** page 36
**Answer: (a) 53 (b) 7 remainder 4**
**Why:** (a) 8 groups of 6 make 48, and 48 + 5 = 53. (b) 53 ÷ 7: 7 × 7 = 49 and 53 - 49 = 4. So 7 remainder 4.

**6** page 36
**Answer: (a) 0, 1, 2, 3, 4 (b) 93**
**Why:** (a) A remainder is always smaller than 5, so it can be 0, 1, 2, 3 or 4. (b) 96 is in the 8 times table, and 96 + 5 = 101 has 3 digits. The next one down: 88 + 5 = 93.

**7** page 36

**Answer: (a) 0, 1, 2, 3, 0, 1, 2, 3 (b) 2 (c) 0**
**Why:** The remainders go 0, 1, 2, 3 and then start again. (b) 30 = 28 + 2, and 28 is in the 4 times table, so the remainder is 2. (c) 100 = 4 × 25, so the remainder is 0.

**8** page 36

**Answer: (a) 31 and 37 (b) 31**
**Why:** Between 30 and 40: 6 × 5 = 30 and 6 × 6 = 36, so the numbers are 31 and 37. Test with 4: 31 ÷ 4 = 7 remainder 3 (right). 37 ÷ 4 = 9 remainder 1 (not right). The number is 31.

**9** page 37

**Answer: (a) 6 (b) 7 (c) 2**
**Why:** 52 ÷ 9 = 5 remainder 7. 5 full buses are not enough, as 7 children are left. So they need 6 buses. The last bus has 7 children, and 9 - 7 = 2 seats are empty.

**10** page 37

**Answer: (a) B (b) D**
**Why:** The pattern repeats every 4 letters. 30 ÷ 4 = 7 remainder 2, so the 30th letter is the 2nd: B. 32 ÷ 4 = 8 with none left, so the 32nd is the last: D.

**11** page 37

**Answer: (a) 77 (b) 8 beads each, 5 left over**
**Why:** (a) 8 × 9 = 72, and 72 + 5 = 77. (b) 77 ÷ 9: 9 × 8 = 72 and 77 - 72 = 5. Each friend gets 8 beads and 5 beads are left over.

**12** page 37

**Answer: 18, 23, 33 and 48**
**Why:** A number leaves remainder 3 when it is 3 more than a number in the 5 times table: 18 = 15 + 3, 23 = 20 + 3, 33 = 30 + 3, 48 = 45 + 3. 41 = 40 + 1 leaves 1.

**13** page 37

**Answer: 8**
**Why:** 47 ÷ 4 = 11 remainder 3. 47 ÷ 5 = 9 remainder 2. 47 ÷ 6 = 7 remainder 5. 47 ÷ 7 = 6 remainder 5. 47 ÷ 8 = 5 remainder 7. The greatest is 7, from 8.

## THINK HARDER

**1** page 38

**Answer: (a) Fri (b) Wed**
**Why:** (a) 45 ÷ 7 = 6 remainder 3. After 6 weeks it is Tuesday again. Then 3 more days: Wednesday, Thursday, Friday. (b) 20 ÷ 7 = 2 remainder 6. Go back 2 weeks to Tuesday, then back 6 more days: Wednesday.

**2** page 38

**Answer: 67**
**Why:** Numbers with remainder 2 when divided by 5 end in 2 or 7. 67 = 66 + 1 (66 is in the 3 times table) and 67 = 64 + 3 (64 is in the 4 times table). No other.

**3** page 38

**Answer: (a) 40 (b) 80 (c) 8**
**Why:** 4 groups of 9 and 4 left make 4 × 9 + 4 = 40. The numbers that work are 10, 20, 30, up to 80: 1 × 10, 2 × 10, ... 8 × 10. A remainder must be less than 9, so 9 remainder 9 is not allowed. The greatest is 80 and there are 8 numbers.

### GO FURTHER

**1** page 39
**Answer: (a) 1 (b) 6 (c) 2**
**Why:** (a) 3 + 5 = 8, and 8 ÷ 7 leaves 1. (b) 3 + 3 = 6, smaller than 7. (c) 5 + 5 = 10, and 10 ÷ 8 leaves 2. Test: 10 + 5 = 15 leaves 1; 20 leaves 6; 26 ÷ 8 leaves 2.

**2** page 39
**Answer: 58**
**Why:** Each clue is 2 short of a full group: 1 left in 3s, 2 left in 4s and 3 left in 5s. So the number is 2 less than a number in the 3, 4 and 5 times tables. The first one is 60, so the number is 58.

**3** page 39
**Answer: Answers vary. Example: I am between 10 and 25. When I am divided by 4 the remainder is 1. When I am divided by 3 the remainder is 2. The only number is 17.**
**Why:** Test each with the second clue. In the example, the numbers with remainder 1 when divided by 4 are 13, 17 and 21. Only 17 leaves remainder 2 when divided by 3.

**4** page 39
**Answer: (a) Yes (b) 1, 2, 4 and 7**
**Why:** Orders he can fill: 3, 5, 6 (3 + 3), 8 (3 + 5), 9, 10, 11 (3 + 3 + 5), 12, 13 (3 + 5 + 5), 14 (3 + 3 + 3 + 5), 15. So the orders he cannot fill are 1, 2, 4 and 7.

### CHECK YOURSELF

**1** page 40
**Answer: 6 remainder 4**
**Why:** 8 × 6 = 48 and 52 - 48 = 4. So 52 ÷ 8 = 6 remainder 4.

**2** page 40
**Answer: 5**
**Why:** 40 ÷ 9 = 4 remainder 4. The 4 people left over need another bus: 5 buses.

**3** page 40
**Answer: 1**
**Why:** 6 × 4 = 24 and 25 - 24 = 1. One muffin is left over.

**4** page 40
**Answer: 29**
**Why:** The numbers from 20 to 30 with remainder 2 when divided by 3 are 20, 23, 26 and 29. Only 29 leaves remainder 4 when divided by 5 (25 + 4).

## Unit 6 · Build a wall

**Your turn** page 41
**Answer: <**
**Why:** 2/3 = 8/12 on the wall. 7/12 is less than 8/12, so 7/12 < 2/3.

### START HERE

**1** page 42
**Answer: (a) 1/3 = 2/6 = 4/12 (b) 2/3 = 4/6 = 8/12 (c) 5/6 = 10/12**
**Why:** Line up the rows. 1 third is the same length as 2 sixths and 4 twelfths. 2 thirds is 4 sixths and 8 twelfths. 5 sixths is 10 twelfths.

**2** page 42
**Answer: (a) 3/8 (b) 4/6 (c) 7/10 (d) 2/5**
**Why:** The bottom number is the number of equal parts in the whole. The top number is the number of shaded parts. A: 3 of 8. B: 4 of 6. C: 7 of 10. D: 2 of 5.

**3** page 42
**Answer: A and C**
**Why:** A fraction needs equal parts. A has 4 equal parts and C has 6 equal sectors. B and D have parts of different sizes.

**4** page 43
**Answer: (a) > (b) > (c) = (d) <**
**Why:** (a) Same bottom number: 5 parts are more than 3. (b) Thirds are larger than quarters. (c) 2/4 = 4/8. (d) 3/4 = 9/12 and 5/6 = 10/12, and 9 is less than 10.

**5** page 43
**Answer: 1/3, 1/2, 2/3, 5/6**
**Why:** Write each one in sixths: 5/6, 2/6, 3/6, 4/6. The order of the tops is 2, 3, 4, 5. So the order is 1/3, 1/2, 2/3, 5/6.

**6** page 43
**Answer: (a) 7 (b) 3 (c) 5 (d) 7**
**Why:** When the bottom numbers are the same, add or subtract the top numbers and keep the bottom number. (d) 1 whole is 10/10, so 10/10 - 3/10 = 7/10.

**7** page 43
**Answer: (a) 3 (b) 3 (c) 5 (d) 2**
**Why:** (a) 1/2 = 2/4, so 2/4 + 1/4 = 3/4. (b) 1/3 = 2/6, so 2/6 + 1/6 = 3/6. (c) 3/4 = 6/8, so 6/8 - 1/8 = 5/8. (d) 1/2 = 3/6, so 5/6 - 3/6 = 2/6.

**8** page 44
**Answer: (a) 7 (b) 5 (c) 1**
**Why:** (a) 3/12 + 4/12 = 7/12. (b) 12/12 - 7/12 = 5/12. (c) 4/12 - 3/12 = 1/12.

**9** page 44
**Answer: (a) 3 (b) 5 (c) 7 (d) D**
**Why:** There are 8 equal steps, so each step is 1/8. A is 3 steps from 0: 3/8. B is 5 steps: 5/8. C is 7 steps: 7/8. D is 4 steps: 4/8, which is half of the line.

**10** page 44
**Answer: (a) 9 (b) 4 (c) 10 (d) 4**
**Why:** Multiply or divide the top and the bottom by the same number. (a) 4 × 3 = 12, so 3 × 3 = 9. (b) 5 × 2 = 10, so 2 × 2 = 4. (c) 1 × 5 = 5, so 2 × 5 = 10. (d) 6 ÷ 2 = 3, so 8 ÷ 2 = 4.

**11** page 44
**Answer: (a) No (b) 5**
**Why:** When the parts are the same size, only the top numbers are added. The bottom number stays 8. 3/8 + 2/8 = 5/8.

## THINK HARDER

**1** page 45
**Answer: 5/12, 6/12 and 7/12**
**Why:** 1/3 = 4/12 and 2/3 = 8/12. The twelfths between 4/12 and 8/12 are 5/12, 6/12 and 7/12. The fractions 4/12 and 8/12 are not between.

**2** page 45
**Answer: (a) 12/16 (b) 18/24**
**Why:** Equal fractions of 3/4 are 3/4, 6/8, 9/12, 12/16, 15/20, 18/24. Their sums are 7, 14, 21, 28, 35, 42. So 12/16 has sum 28 and 18/24 has sum 42.

**3** page 45
**Answer: (a) 7 (b) 5**
**Why:** 1/4 = 3/12 and 1/3 = 4/12. (a) 3/12 + 4/12 = 7/12. (b) 12/12 - 7/12 = 5/12.

### GO FURTHER

**1** page 46
**Answer: (a) 4 (b) 3 (c) 4**
**Why:** (a) 3/4 - 1/2 = 3/4 - 2/4 = 1/4. (b) 5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3. (c) 7/12 - 1/3 = 7/12 - 4/12 = 3/12 = 1/4.

**2** page 46
**Answer: (a) 0 halves, 2 quarters, 4 eighths**
**Why:** In eighths a half is 4, a quarter is 2 and an eighth is 1. 6 pieces must make 8 eighths. 2 quarters (4) and 4 eighths (4) make 8. A half would leave 5 pieces to make 4 eighths, which cannot be done.

**3** page 46
**Answer: (a) 2 (b) 6**
**Why:** 1/2 of the bar is 6 parts and 1/3 of the bar is 4 parts. 6 + 4 = 10 parts are coloured, so 2 parts are white. 2/12 is the same as 1/6.

**4** page 46
**Answer: Answers vary. Example: Ali eats 3/8 of a pizza and Mei eats 2/8 of it. Together they eat 5/8 of the pizza.**
**Why:** In the example, 3/8 + 2/8 = 5/8.

### CHECK YOURSELF

**1** page 47
**Answer: 5**
**Why:** The bar has 8 equal parts and 5 are shaded. The fraction is 5/8.

**2** page 47
**Answer: <**
**Why:** 3/4 = 6/8. 5/8 is less than 6/8, so 5/8 < 3/4.

**3** page 47
**Answer: 3**
**Why:** 1/4 = 2/8, so 5/8 - 2/8 = 3/8.

**4** page 47
**Answer: 8**
**Why:** 3 × 4 = 12, so 2 × 4 = 8. 2/3 = 8/12.

## Unit 7 · Around and inside

**Your turn** page 48
**Answer: Area 36 cm². Perimeter 26 cm.**
**Why:** Area: 9 × 4 = 36 cm². Perimeter: 9 + 4 + 9 + 4 = 26 cm, or 2 × (9 + 4) = 26 cm.

### START HERE

**1** page 49
**Answer: (a) A: 8 cm², 12 cm (b) B: 6 cm², 14 cm (c) C: 6 cm², 10 cm (d) B,C**
**Why:** Area is the number of squares: A 8, B 6, C 6. A is 2 × (4 + 2) = 12 and C is 2 × (3 + 2) = 10. B fits a 4 by 2 box (12), and the dent adds 1 cm on each side: 14. B and C have the same area but different perimeters.

**2** page 49
**Answer: (a) Area 48 cm², perimeter 28 cm (b) Area 44 m², perimeter 30 m**
**Why:** (a) Area 8 × 6 = 48. Perimeter 2 × (8 + 6) = 28. (b) Area 11 × 4 = 44. Perimeter 2 × (11 + 4) = 30.

**3** page 49
**Answer: (a) 28 (b) 49 (c) 9**
**Why:** (a) 4 × 7 = 28. (b) 7 × 7 = 49. (c) 4 equal sides: 36 ÷ 4 = 9 m.

**4** page 49
**Answer: (a) 6 (b) 54**
**Why:** Length + width is half of the perimeter: 30 ÷ 2 = 15. So the width is 15 - 9 = 6 cm. The area is 9 × 6 = 54 cm².

**5** page 50
**Answer: (a) 5 (b) 3 (c) 15**
**Why:** (a) 15 ÷ 3 = 5. (b) 9 ÷ 3 = 3. (c) 5 rows of 3 tiles make 5 × 3 = 15 tiles.

**6** page 50
**Answer: (a) 3 (b) 12**
**Why:** Length + width = 14 ÷ 2 = 7. The pairs are 6 and 1 (area 6), 5 and 2 (area 10), 4 and 3 (area 12). That is 3 rectangles. The one closest to a square, 4 by 3, has the greatest area: 12 cm².

**7** page 50
**Answer: (a) 4 (b) 3 (c) 36 (d) 28**
**Why:** a = 8 - 4 = 4 cm and b = 6 - 3 = 3 cm. Area: a 4 by 6 rectangle and a 4 by 3 rectangle: 24 + 12 = 36 cm². Perimeter: push the steps out to an 8 by 6 box: 2 × (8 + 6) = 28 cm.

**8** page 51
**Answer: (a) 12 (b) 18 (c) 6 (d) 18**
**Why:** (a) 6 + 4 + 2 = 12 cm². (b) Same as its 6 by 3 box: 2 × (6 + 3) = 18 cm. (c) The rectangle has 18 squares: 18 - 12 = 6 more. (d) 18 cm: filling the steps keeps the perimeter.

**9** page 51
**Answer: (a) 6 (b) Ben's (c) 16**
**Why:** (a) Mei's perimeter is 2 × (10 + 2) = 24 cm, so Ben's side is 24 ÷ 4 = 6 cm. (b) Ben 6 × 6 = 36 cm², Mei 10 × 2 = 20 cm². (c) 36 - 20 = 16 cm².

**10** page 51
**Answer: (a) A and B (b) C**
**Why:** A square that touches the figure on only 1 side adds 2 to the perimeter: A and B give 12 cm. C fits into the corner and touches 2 sides: it hides 2 cm and adds 2 cm, so the perimeter stays 10 cm.

**11** page 51
**Answer: (a) 8 (b) 32**
**Why:** (a) 8 × 8 = 64, so each side is 8 m. (b) The border is the perimeter: 32 m.

## THINK HARDER

**1** page 52
**Answer: (a) 24 (b) 48 (c) 16**
**Why:** Each rectangle is 8 cm by 4 cm: perimeter 24 cm, so 48 cm in all. The square has 32 cm, and 48 - 32 = 16. The 8 cm cut is on both pieces: 16 cm.

**2** page 52
**Answer: (a) 9 (b) 4**
**Why:** Length + width is 26 ÷ 2 = 13. The pairs that multiply to 36: 1 and 36, 2 and 18, 3 and 12, 4 and 9, 6 and 6. The pair that adds up to 13 is 4 and 9.

**3**
page 52

**Answer: (a) 25 (b) 22 (c) 22**
**Why:** (a) 16 + 9 = 25 cm². (b) No dents, so the perimeter equals its 7 cm by 4 cm box: 2 × (7 + 4) = 22 cm. (c) A 4 by 7 box and no dents: 22 cm again.

### GO FURTHER

**1**
page 53

**Answer: (a) 22 (b) 14**
**Why:** Each new square adds 2 cm (1 cm on top and 1 cm below). The perimeter is 4 + 2 for every extra square. For 10 squares: 4 + 9 × 2 = 22 cm. For 30 cm: (30 - 4) ÷ 2 = 13 extra squares, so 14 squares.

**2**
page 53

**Answer: (a) 8 m by 6 m (b) 24**
**Why:** The path adds 1 m on each side, so both lengths grow by 2 m: 6 + 2 = 8 and 4 + 2 = 6. The whole rectangle has an area of 8 × 6 = 48 m². The pond has 6 × 4 = 24 m². The path is 48 - 24 = 24 m².

**3**
page 53

**Answer: (a) 2 (b) 30**
**Why:** The side must divide both 12 and 10. 1 and 2 do; 3, 4, 5 and 6 each fail one of them. So the largest side is 2 m: 6 tiles along and 5 across, 30 tiles.

### CHECK YOURSELF

**1**
page 54

**Answer: 45**
**Why:** Area is length × width: 9 × 5 = 45 cm².

**2**
page 54

**Answer: 5**
**Why:** A square has 4 equal sides: 20 ÷ 4 = 5 cm.

**3**
page 54

**Answer: Area 7 cm². Perimeter 12 cm.**
**Why:** Area: 3 + 2 + 2 = 7 cm². Perimeter: the figure fits in a 3 by 3 box and has no dents on the outside, so 2 × (3 + 3) = 12 cm.

**4**
page 54

**Answer: 7**
**Why:** Length + width is 22 ÷ 2 = 11. The length is 11 - 4 = 7 cm.

## Unit 8 · Corners and lines

**Your turn**
page 55

**Answer: smaller than**
**Why:** The second line is inside the corner of the checker, so the angle is smaller than a right angle.

### START HERE

**1**
page 56

**Answer: (a) B and E (b) A and F (c) C and D**
**Why:** A and F fit the checker exactly, so they are right angles. B and E are inside the corner of the checker, so they are smaller. C and D stick out past the corner of the checker, so they are greater.

**2**
page 56

**Answer: (a) 5 (b) 6 (c) 2**
**Why:** A is an L: 5 right angles, and its dent is greater than a right angle. B has 6 right angles and 2 dents. C has right angles only at its 2 left corners; its 2 sloping corners do not fit the checker.

**3**
page 56

**Answer: (a) F and N (b) F and L**
**Why:** F has parallel arms and a stem at right angles to them. V has neither. N has 2 parallel upright lines, but its sloping line is not at a right angle. L has 2 lines that meet at a right angle.

**4**
page 57

**Answer: (a) 3 (b) 2 (c) 1 (d) 6**
**Why:** There are 3 small angles. The next size is 2 small angles side by side: 2 of those. Then 1 angle made of all 3 small ones. 3 + 2 + 1 = 6 angles.

**5**
page 57

**Answer: (a) R (b) P**
**Why:** AB goes 4 across and 2 up. R goes 4 across and 2 up too, so it never meets AB: parallel. P goes 2 across and 4 down, a right angle to AB: perpendicular.

**6**
page 57

**Answer: (a) 4 (b) 12**
**Why:** At each end of a rung, the rung and the rail make 2 right angles (above and below the rung). A rung has 2 ends: 4 right angles. 3 rungs: 3 × 4 = 12.

**7**
page 58

**Answer: (a) E (b) W (c) S**
**Why:** Turning right from north: east, south, west. (a) 1 turn: east. (b) 3 turns: west. (c) 2 turns left: west, then south. Two right angles either way: south.

**8**
page 58

**Answer: (a) 4 (b) 2 (c) 1 (d) perpendicular**
**Why:** (a) Each corner of a rectangle is a right angle: 4. (b) The 2 long sides are parallel and the 2 short sides are parallel: 2 pairs. (c) 2 right angles would make 2 parallel sides, which never meet to close a triangle: at most 1. (d) The sides of a square meet at right angles.

**9**
page 58

**Answer: (a) True (b) False (c) True (d) False**
**Why:** (a) That is what perpendicular means. (b) Parallel lines never meet. (c) A line across the 2 rails of a ladder is perpendicular to both. (d) Lines can cross at any angle.

**10**
page 58

**Answer: (a) No (b) greater than**
**Why:** Angle X is a dent. It points into the shape. A checker does not fit inside it: the angle is greater than a right angle.

## THINK HARDER

**1**
page 59

**Answer: (a) 10 (b) 21**
**Why:** With 5 lines there are 4 small angles. Then 3 angles made of 2 small angles, 2 made of 3, and 1 made of 4. 4 + 3 + 2 + 1 = 10. With 7 lines the small angles are 6: 6 + 5 + 4 + 3 + 2 + 1 = 21.

**2**
page 59

**Answer: (a) R**
**Why:** AB goes 2 across for every 1 up. A perpendicular line goes 1 across and 2 down. From C, keep going 1 across and 2 down until you reach the bottom row: R.

**3**
page 59

**Answer: (a) 8 (b) 4 (c) 7**
**Why:** The plus has 12 corners. 4 of them dent into the shape: they are greater than a right angle. The other 8 fit a checker. In (c) the 10 corners are right angles or dents. 10 - 3 = 7 right angles.

### GO FURTHER

**1**
page 60

**Answer: (a) 180 (b) 360 (c) 45 (d) 270**
**Why:** (a) 90 + 90 = 180. (b) 4 × 90 = 360. (c) 90 ÷ 2 = 45. (d) 3 × 90 = 270.

**2**
page 60

**Answer: (a) parallel (b) Yes**
**Why:** (a) Two lines that are perpendicular to the same line stay the same distance apart and never meet: they are parallel. (b) A line across 2 parallel lines makes the same angle with both, so yes.

**3**
page 60

**Answer: (a) 2 (b) 16 (c) 36**
**Why:** (a) 2 lines across and 2 lines up: 2 pairs. (b) 2 × 2 = 4 crossings, each with 4 right angles: 16. (c) 3 × 3 = 9 crossings, 9 × 4 = 36 right angles.

### CHECK YOURSELF

**1**
page 61

**Answer: greater than**
**Why:** The second line sticks out past the corner of the checker: the angle is greater than a right angle.

**2**
page 61

**Answer: E and H**
**Why:** E has 3 parallel arms. H has 2 parallel upright lines. V has 2 sloping lines that are not parallel. T has no pair of parallel lines.

**3**
page 61

**Answer: 2**
**Why:** The 2 long sides are parallel and the 2 short sides are parallel.

**4**
page 61

**Answer: N**
**Why:** Turning left from east, the next direction is north.

## Unit 9 · Measure it, pay it

**Your turn**
page 62

**Answer: 1 ℓ 550 ml**
**Why:** 2 ℓ 350 ml = 2350 ml. Then 2350 - 800 = 1550 ml = 1 ℓ 550 ml. Check: 1550 + 800 = 2350.

### START HERE

**1**
page 63

**Answer: (a) 300 (b) 4050 (c) 5300 (d) 2 kg 750 g (e) 3 m 65 cm**
**Why:** (a) 3 × 100 = 300. (b) 4 kg = 4000 g, 4000 + 50 = 4050. (c) 5 ℓ = 5000 ml, 5000 + 300 = 5300. (d) 2000 g is 2 kg, 750 g left. (e) 300 cm is 3 m, 65 cm left.

**2**
page 63

**Answer: (a) 8 (b) 7 (c) 1**
**Why:** The pencil starts at 3 and ends at 11: 11 - 3 = 8 cm. The crayon starts at 6 and ends at 13: 13 - 6 = 7 cm. The pencil is 8 - 7 = 1 cm longer.

**3** page 63
**Answer:** C, A, D, B
**Why:** Write each in grams: A is 1080 g, B is 1900 g, C is 1008 g and D is 1090 g. In order: 1008, 1080, 1090, 1900. That is C, A, D, B.

**4** page 63
**Answer: (a)** 6 **(b)** 200
**Why:** 3 ℓ 200 ml = 3200 ml. 3200 ÷ 500 = 6 remainder 200. She fills 6 pots, with 200 ml left.

**5** page 64
**Answer: (a)** 350 **(b)** 600 **(c)** 850 **(d)** 500
**Why:** Each small mark is 50 ml. A is 1 mark above 300: 350 ml. B is at 600 ml. C is 1 mark above 800: 850 ml. 850 - 350 = 500 ml.

**6** page 64
**Answer: (a)** 4.05 **(b)** 0.95
**Why:** (a) 2.65 + 1.40 = 4.05. (b) 5.00 - 4.05 = 0.95.

**7** page 64
**Answer: (a)** 14.45 **(b)** 5.55 **(c)** Yes
**Why:** (a) 6.20 + 4.85 = 11.05, and 11.05 + 3.40 = 14.45. (b) 20.00 - 14.45 = 5.55. (c) 14.45 + 1.75 = 16.20, which is less than $20. So she can.

**8** page 64
**Answer:** 1100
**Why:** 2 kg 400 g = 2400 g. 1 melon: 2400 ÷ 3 = 800 g. 2 apples: 2 × 150 = 300 g. Together: 800 + 300 = 1100 g.

**9** page 65
**Answer: (a)** 06:15 **(b)** 13:40 **(c)** 21:05 **(d)** 4:50 **(e)** 8:25
**Why:** After 12 noon, add 12 to the hour for 24-hour time. (b) 1 + 12 = 13. (c) 9 + 12 = 21. For (d) and (e), take away 12: 16 - 12 = 4 and 20 - 12 = 8.

**10** page 65
**Answer: (a)** 1 h 15 min **(b)** 4:35 pm **(c)** 6:30 pm
**Why:** (a) 5:25 to 6:00 is 35 min, then 6:00 to 6:40 is 40 min. 35 + 40 = 75 min = 1 h 15 min. (b) 3:40 + 20 min = 4:00, then 35 more min = 4:35 pm. (c) 8:15 pm - 1 h = 7:15 pm, then 45 min back = 6:30 pm.

**11** page 65
**Answer: (a)** 4:10 **(b)** 7:45 **(c)** 11:25
**Why:** The short hand shows the hour, the long hand the minutes. A: long hand at 2 is 10 minutes, short hand just past 4. B: long hand at 9 is 45 minutes. C: at 5 is 25 minutes.

**12** page 65
**Answer: (a)** 5.20 **(b)** 4.80
**Why:** (a) 8 × 65¢ = 520¢ = $5.20. (b) 10.00 - 5.20 = 4.80.

## THINK HARDER

**1** page 66
**Answer:** 11:00
**Why:** Lesson 1: 07:30 to 08:15. Break. Lesson 2: 08:25 to 09:10. Break. Lesson 3: 09:20 to 10:05. Break. Lesson 4: 10:15 to 11:00. The 4th lesson ends at 11:00.

**2** page 66
**Answer:** 1 h 20 min
**Why:** From 18:35 to 19:00 is 25 min. 19:00 to 20:00 is 1 h. 20:00 to 20:10 is 10 min. The whole time is 1 h 35 min. Take away the 15 min break: 1 h 20 min.

**3** page 66
**Answer: (a) 1.20 (b) 2.70**
**Why:** The 2 notebooks cost $3 more than 2 pens (2 × $1.50). Take that away: $9 - $3 = $6 for 5 pens. A pen costs $6 ÷ 5 = $1.20. A notebook costs $1.20 + $1.50 = $2.70.

### GO FURTHER

**1** page 67
**Answer: (a) 2 (b) 4**
**Why:** (a) The 3 ℓ jug takes 3 ℓ, so 5 - 3 = 2 ℓ are left in the 5 ℓ jug. (b) The 3 ℓ jug holds 2 ℓ and has room for 1 ℓ more. The full 5 ℓ jug gives 1 ℓ, so 5 - 1 = 4 ℓ are left. That is exactly 4 ℓ.

**2** page 67
**Answer: (a) 4:28 pm (b) 4:37 pm**
**Why:** (a) A slow watch is behind: the real time is 8 minutes later: 4:20 pm + 8 min = 4:28 pm. (b) At the real time 4:45 pm her watch shows 8 minutes earlier: 4:37 pm.

**3** page 67
**Answer: Answers vary. Example: A book costs $6.40 and a pen costs $2.15. How much more does the book cost than the pen? The answer is $4.25.**
**Why:** The only way is the difference: 6.40 - 2.15 = 4.25. A sum of the two would be $8.55.

**4** page 67
**Answer: 07:15**
**Why:** From 22:30 to 24:00 (midnight) is 1 h 30 min. The rest of the shift is 8 h 45 min - 1 h 30 min = 7 h 15 min after midnight. So it ends at 07:15.

### CHECK YOURSELF

**1** page 68
**Answer: 240**
**Why:** 2 m = 200 cm, and 200 + 40 = 240 cm.

**2** page 68
**Answer: 500**
**Why:** 1 ℓ 200 ml = 1200 ml. Then 1200 - 700 = 500 ml.

**3** page 68
**Answer: 1 h 30 min**
**Why:** 9:40 to 10:00 is 20 min. 10:00 to 11:00 is 1 h. 11:00 to 11:10 is 10 min. In all: 1 h 30 min.

**4** page 68
**Answer: 2.65**
**Why:** 5.00 - 2.35 = 2.65.

## Unit 10 · Think it through

**Your turn** page 69
**Answer: 11**
**Why:** 2 children are in front of Lily and 8 are behind her. 2 + 1 + 8 = 11. Check: 3 + 9 - 1 = 11.

### START HERE

**1** page 70
**Answer: (a) Mei (b) 15 (c) 25 (d) 105**
**Why:** (a) The tallest bar is Mei's. (b) 30 - 15 = 15. (c) Siti read 10 books, so Kumar read 10 + 15 = 25. (d) 25 + 15 + 30 + 10 + 25 = 105.

**2** page 70

**Answer: (a) 70 (b) 110 (c) 90 (d) 270**
**Why:** The labelled lines go up in 20s and a thin line is halfway between, so each thin line is 10. A is 70, B is 110, C is 90. Together: 70 + 110 + 90 = 270.

**3** page 70

**Answer: (a) Tue (b) Mon**
**Why:** (a) 3rd to 24th is 21 days = 3 weeks. So the 24th is a Tuesday again. (b) 3rd to 30th is 27 days = 3 weeks and 6 days. 6 days after Tuesday is Monday.

**4** page 71

**Answer: (a) grapes (b) apple (c) banana (d) orange**
**Why:** Only Lily has a 4-letter name: orange. Kumar has neither apple nor banana: grapes. Wei does not have the apple: banana. Aisha has the apple.

**5** page 71

**Answer: (a) 9 (b) 10**
**Why:** (a) Ravi is 3rd. The 5 children between are 4th to 8th. Mei is 9th and last. (b) 7 children are in front of Aisha and 2 are behind her: 7 + 1 + 2 = 10.

**6** page 71

**Answer: (a) milk bun, milk pie, milk fruit, tea bun, tea pie, tea fruit (b) 3 (c) 4**
**Why:** Fix the drink and list the snacks in order: milk bun, milk pie, milk fruit, then tea bun, tea pie, tea fruit. 3 meals have milk. A bun or a pie: milk bun, milk pie, tea bun, tea pie, so 4.

**7** page 72

**Answer: (a) 31 (b) 8**
**Why:** Each new hexagon adds 5 sticks: 1: 6, 2: 11, 3: 16, 4: 21, 5: 26, 6: 31. For 41 sticks, count on: 7 hexagons need 36 and 8 hexagons need 41.

**8** page 72

**Answer: 162, 252 and 342**
**Why:** The hundreds digit is 1, 2 or 3. The ones digit is 2, 4 or 6 and the tens digit is greater. With 1: tens + ones = 8, so 6 and 2. With 2: 7, so 5 and 2. With 3: 6, so 4 and 2. The numbers are 162, 252 and 342.

**9** page 72

**Answer: 7**
**Why:** Work backwards. Add 4 to 17: 21. Divide by 3: 7.

**10** page 72

**Answer: 2.90**
**Why:** 2 juices cost 2 × 1.85 = 3.70. With the sandwich: 3.70 + 3.40 = 7.10. Change: 10.00 - 7.10 = 2.90.

## THINK HARDER

**1** page 73

**Answer: 18**
**Why:** Height 1: width 1 gives 6, width 2 gives 4 (2 in each row), width 3 gives 2. Height 2: width 1 gives 3, width 2 gives 2, width 3 gives 1. In all: 6 + 4 + 2 + 3 + 2 + 1 = 18.

**2** page 73

**Answer: Aisha**
**Why:** If Ben did it, Aisha and Wei both tell the truth: 2 truths. If Wei did it, Ben and Aisha both tell the truth: 2 truths. If Aisha did it, only Wei tells the truth: exactly 1. So Aisha did it.

**3**
page 73

**Answer: (a) 30 (b) 45 (c) 60 (d) 50**
**Why:** Monday is 1 box, Tuesday 1 box and 15, Wednesday 2 boxes, Thursday 2 boxes take away 10. Total: 6 boxes + 5 = 185, so 1 box = 30. Monday 30, Tuesday 45, Wednesday 60, Thursday 50.

## GO FURTHER

**1**
page 74

**Answer: (a) 16 (b) 9 (c) 4 (d) 30**
**Why:** There are 16 small squares. A 2 by 2 square fits in 3 places across and 3 down: 9. A 3 by 3 square fits 2 across and 2 down: 4. There is 1 big square. 16 + 9 + 4 + 1 = 30.

**2**
page 74

**Answer: (a) 1 (b) 2**
**Why:** (a) Put 1 coin on each side. If the scale tips, the light coin is on the high side. If it balances, it is the coin left out: 1 weighing. (b) Weigh 3 coins against 3 coins to find the group of 3 with the light coin. Then use (a): 2 weighings.

**3**
page 74

**Answer: (a) 1 (b) 7**
**Why:** If all 8 coins were 10¢, they would be worth 80¢. They are worth 150¢, 70¢ more. Changing one 10¢ coin to a 20¢ coin adds 10¢, so 7 coins are 20¢ and 1 coin is 10¢.

**4**
page 74

**Answer: Answers vary. Example: Lily, Wei and Kumar each play a different game: chess, ball or cards. Lily does not play ball. Wei plays neither chess nor cards. Kumar does not play chess. Answer: Lily plays chess, Wei plays ball, Kumar plays cards.**
**Why:** In the example, Wei plays ball (clue 2), then Kumar cannot play chess (clue 3), so Kumar plays cards and Lily plays chess.

## CHECK YOURSELF

**1**
page 75

**Answer: 30**
**Why:** Monday is at 20 and Tuesday is at 50. 50 - 20 = 30.

**2**
page 75

**Answer: 9**
**Why:** 2 + 1 + 6 = 9. Do not forget Ben.

**3**
page 75

**Answer: Ben**
**Why:** From the shortest: Lily, Mei, Ben, Aisha. The tallest is Aisha, so the 2nd tallest is Ben.

**4**
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**Answer: 30**
**Why:** The numbers go up by 4. The 8th number is 2 + 7 × 4 = 30.